

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

The release of this audit program to parties external to IBM requires the written approval of the Chief Auditor.

This program is intended as a guide for assessing risk management in this process through the performance of an audit or a similar engagement. Risk management is the process to identify, manage, and control potential events to provide reasonable assurance related to the achievement of IBM's business and control objectives. IBM's control objectives fall into five broad categories:

1. compliance with external laws and regulations, as well as IBM policies and procedures
2. accomplishment of business objectives
3. reliable information for financial reporting and decision making
4. efficient operations, and
5. safeguarding both intellectual and physical assets.

The control assessment questions are intended to address process objectives from the categories above and the risks to achieving those objectives. Objectives and risks in the actual process under review may differ based on organizational, operational, regulatory, and economic factors. As such, individuals performing the engagement must formulate a test strategy based on risks relative to the actual process under review, regardless of whether the risks are identified within this program.

DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
June 2025	All	All	Jen Daniels	Updated TOC; aligned Key Control Questions/Tests to RAE including renumbering tests; updated list of tools; added reference to CAAS data privacy programs
May 2024	Process & Obj.	Audit Program Risk Alignment SD006.P – CC / RFS SE004.P – CC / RFS Reference material	Wolfgang Stengel	Adding separate risk area for data security and privacy; Updating control points accordingly (adding SD006.P, changing SE004.P); Updating links
April 2024	Process & Obj. Solution Design Solution Delivery	Reference Material Solution Design Solution Delivery SE002.P - Delivery SE006.P - Labor SE008.S - SoD	Wolfgang Stengel	Multiple minor updates reflecting changes to IDR
April 2023	2. Services Solution Delivery	Data Security and Privacy (SE010.P)	Don Dobrzelewski	Added Privacy Testing
February 2023	All	All	CA&AS	Converted from Notes into a Word document and into a Box folder

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PROCESS AND OBJECTIVE

High-Level Overview

- The Solution Design and Delivery (SDD) processes starts with due diligence activity to understand customer requirements and the scope of work, from which a solution design is developed.
- That solution is then priced, and terms and conditions for its delivery defined to best manage the risks relating to the solution.
- Approvals at this stage will be dependent on the nature of the solution from a business perspective, the financial value (expected revenue and profitability) and if there are any variations to standard terms and conditions requiring review by the relevant subject matter experts (SME) within IBM.
- Having confirmed the proposed solution, price and contract terms within IBM, a formal proposal would be presented to the customer. On many of IBM's larger signings, the negotiation phase can last many months and be iterative with aspects of the proposal being amended during the negotiation stage.
- A key factor during this phase is to assess whether any commitments have been made during negotiation without the required approvals. Also, if changes made during negotiation and solution design require additional reviews or approvals.
- Having signed the contract, the solution delivery phase starts with the implementation of actions and services in accordance with the contract. Often contracts will have a transition phase where services are transitioned to IBM followed by a transformation stage (T&T). Service delivery and service level performance for which IBM is held responsible for during these phases is monitored against contractual requirements.

SDD processes are relevant for new business ranging from low value additional business with an existing customer under the terms of an existing contract (often referred to as requests for service, RFS) to new business with a new customer for services that IBM has not delivered previously. The requirements in the SDD processes are defined based upon the risk associated with the many variations of new business that IBM can engage in.

Key Considerations

- Complexity of the audit: features of more complex engagements can include the following: cross-brand engagements, portfolio audits with many contracts, contracts with financial and service level issues, first of kind solutions or extended transition and transformation phases.
- The audit should determine if an effective system of controls is in place at strategic, tactical, and operational levels, and how this combines to best manage risk. In addition to a review of the mandatory activities required in the SDD processes, the quality of work performed should be assessed to ensure that customer expectations are met, whilst delivering the solution in accordance with IBM policies, guidelines, and instructions.
- Initial data analysis of a contract or portfolio of contracts should focus on assessing where material variances to financial and service level targets exist and identifying those areas within the SDD processes that can be the root cause of such issues. This should shape the focus of process reviews and testing.
- Consideration should be given to other review activities performed (such as by business controls) of the contract/s. Focus the audit review on the period after those reviews and also assess actions taken to address review recommendations.
- Analysis should assess which aspects of the SDD processes present the highest risk on the contract/s and establish the root cause of issues (as an example, are missed service levels due to poorly performed solution design whereby inappropriate service levels were agreed).

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PROCESS AND OBJECTIVE (Continued)

- The time from when the contract, contract change or request for service was signed will typically govern whether the area of the CRMfS audit should be more focused towards Solution Design or more towards Solution Delivery. As a general guidance, the following can help determine this:
 - Look at when the main contract was signed. If the contract involves Transition and Transformation, then the ideal time to review Solution Design is when Transition has completed, and Transformation is well underway. At this time, the auditor will be able to identify if the contract is starting to deviate from the approved business case and should try to ascertain the root cause(s) of that deviation. This would generally be 12-18 months after signature. In other cases, where the main contract was more recently signed, the auditor would typically test Solution Design by reviewing any pre-contract reviews, identifying any risks called out in these reviews, and establishing whether identified risks had been mitigated prior to signature or as part of the transition or transformation phases.
 - If the main contract has been running for a longer period, then the Solution Delivery aspect will most likely require more focus. However, Solution Design may still be relevant to review when more recent contract changes or subsequent requests for service have occurred.

Process Interfaces

- Although the CRMfS audit program is focused on the customer solution and the customer facing engagement, the processes include activities relating to asset management, procurement, accounts payable, accounting, signing and backlog and sales transaction support (fulfilment) that are typically delivered by centers of excellence for multiple contracts, countries, and lines of business.
- Each of these processes has an associated audit program, and careful consideration should be given to determine the nature and extent of any testing performed of these processes to avoid extending the scope from CRMfS into other areas and duplicating audit effort.
- The line between one audit program and another is not strictly defined, and the extent of testing should depend on the auditor's assessment of the risks on the contract/s under review. It is important to bear this in mind when scoping audits in this area, to avoid any potential overlap with deeper process audits of Fulfilment.

A key consideration when reviewing SDD is to assess the efficient and effective coordination of all involved organizations needed in order to meet the business objectives (both during design and delivery). The auditor should be aware of each organization's responsibilities and associated risks, both during the design and delivery phases of work.

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PLANNING: Audit Program Risk Alignment

Customer Relationship Management for Services

Defined Risk Area	Audit Risk Level	Key Control Question	Test Number
Solution Design	High	Services Solution Design	SD001.P, SD002.P, SD003.P SD004.S, D005.P
Solution Delivery	High	Services Solution Delivery	SE001.P, SE002.P, SE003.P, SE004.P, SE005.P, SE006.P, SE007.S, SE008.S, SE010.P, SE011.P
Data Security and Privacy	High	Data Protection and Privacy	DS001.P
Financial Management	High	Financial Management	FM001.P, SF002.P
Labor Claiming	Medium	Labor Claiming	LC001.P
Resource Management	Medium	Resource Management	RM001.P

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PLANNING: Data Analysis

The complexity of CRMfS engagements requires robust and thorough data analysis to support a risk-based auditing approach. In addition, a thorough understanding of processes operating at strategic, tactical, and operational levels is required, and data analysis should not purely focus on the more traditional transaction level of data. Support for analysis can be sought from the Internal Audit Technology and Support team.

Reconciliation of data

Data used for analysis should be reconciled to ensure that universe information is complete. A review of complementary data should be performed (e.g., comparing invoices and accrual data with recorded revenue) to confirm that consistent assumptions have been used for the data provided. Where possible, reconciliation to an independent source should be performed.

The [Data Analysis engagement procedures in the Audit Manual](#) (link internal to CA&AS members only) provide further suggestions.

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PLANNING: Data Analysis (Continued)

The table below summarizes the key data that should be analyzed to identify areas of focus for a CRMfS audit:

Data to be Analyzed	Analysis to be Performed	Possible Concerns
Contract Selection (portfolio audits)	A sample of contracts for review during the audit should be selected with consideration for the following factors: • Total contract value • Complexity • Negative, low, inconsistent (with the rest of the portfolio) GP • Significant variances to pricing case • Quality assurance type issues documented contract reviews. • Contracts in a continued and ongoing troubled / underperforming status • Customer satisfaction issues, material claims/settlements and complaints • Contracts terminated early or reduced in scope • Financial indicators of risk including: • High amounts of aged debt, and disputed invoices • High value of write offs • High value and number of subcontracted services/suppliers In addition, the sample should aim to be representative: • include contracts that are newly signed (to assess solution design issues) and longer running contracts • include contracts by sector, country Selected samples should not just be restricted to troubled contracts.	• Inconsistencies between measurement data e.g., poor financial and delivery performance with high rating assessments. Indicates incomplete oversight on risk areas within a contract portfolio. • Completeness of universe, and integrity of data fields in the report, that provide incomplete information to management limiting visibility to issues • Inception to date GP less than planned/priced GP (contracts requiring significant recovery to meet planned GP) / downward trends in GP, as an indicator of transition and transformation issues, or insufficient action to manage contract recovery High levels of customer satisfaction, that masking unauthorized commitments / settlements

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Possible Concerns
Overall Business Performance Measurements	Review measurements and assess the impact of: • Revenue and GP trends • Signings and backlog trends and significant backlog adjustments • Resource utilization (typically in IBM Consulting) • Compliance testing and key control (SOX) testing results	• Significant variances from revenue and GP plans, may indicate changing / unplanned changes in assumptions on the contract. • deteriorating GP trends, and scope and service increases without associated revenue may indicate contract clauses that commit IBM to unquantified risks (known as sweeper clauses). • Declining signings and backlog could lead to high-risk business behavior. As example could be signing contracts which are likely to be problematic to delivery against or less advantageous to IBM. • Material variances in resource utilization versus target (IBM Consulting), and anomalous utilization levels, may indicate manipulation of business performance. • Compliance testing results that are inconsistent with other indicators may hide the visibility of issues to senior management • Issues in compliance testing may highlight concerns by independent reviewers that are not recognized as an issue by local management
Revenue Recognition	Review ledger entries report paying attention to: • Material manual revenue and cost deferrals and accruals • Revenue reversals, negative revenue in the month following a quarter end • Manual credits and provisions • Contracts with negative revenue in the month following quarter end	• Unsupported accounting entries, with insufficient review by either management or Accounting resulting in inaccurate or overstated revenue or GP • Inappropriate actions to overstate revenue at period end, resulting in revenue / GP overstatement that impact incentive or commission payments
Claims and customer disputes	Review claims, claims avoidance, settlements, write offs: • High value claims and settlements • Claims and settlements recorded after the announcement of the audit • High frequency of claims	• Continued and frequent claims may indicate inaction to resolve the root cause of issues or concessions. • Consideration should be given to other indicators of settlements without or prior to appropriate approvals (and outside of the claims process)

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Possible Concerns
Work at Risk/Work prior to contract signature	Review the management system in place to track and control work at risk and pre-contract work. In some business lines (in particular IBM Consulting) pre-contract work is often used as part of the standard operating process on contracts to improve utilization and customer satisfaction within tolerated limits. - Identify reported work at risk and agreed pre-contract work. - Identify labor claiming prior to those approvals - Identify conversion rates of pre-contract work to signed deals. - Assess whether IBM is requiring subcontractors to work at risk for IBM - Review instances for work at risk or pre-contract work extensions	- Assess the amount of work at risk or pre-contract work that resulted in unbilled revenue / unrecoverable costs. - Assess whether work at risk or pre-contract work was effectively identified, monitored, and controlled so that it did not exceed approved limits and identified unreported work at risk. - Assess whether IBM was adequately protected, and whether there was limitation of liability in instances of work at risk and pre-contract work. - Focus analysis to identify first instances of work at risk (in particular on portfolio reviews) - Focus analysis to identify higher risk instances of work at risk (e.g., higher risk with work associated to access to customer equipment versus provision of education). - Assess whether the need to work at risk or perform pre-contract work indicates other process failures in IBM or whether issues are customer driven.
Billing and Accounts Receivable	Review credit notes, including: - Manual credits - Credits for higher risk reason codes - Credits processed near quarter end periods - High-value credits - Repeat credits	- High volumes of manual credits may indicate process issues with billing accuracy, and review and validation of invoicing - Assess whether credits are in accordance with approval requirements (including independent review outside of the business line where required) - Consideration should be given to whether credits were correctly classified (by reason code) and therefore whether correct approvals were sought. - Assess whether credits should have been approved as claims, or indicate unauthorized commitments - Assess whether credits indicate ongoing issues that are not being resolved.

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Possible Concerns
Procurement	Review purchase order and payment data: - Review invoices with purchase orders raised close to or after invoice dates. Review report of bypasses committed and identify repeat offenders: - Assess the actions taken by management to address repeat offenders	- Lack of Procurement involvement may result in increased costs to IBM, and indicate instances where "arm's length" relationships are not in place - Continued bypass activity may indicate lack of monitoring of spend on a contract - Where reliance is placed on third parties for the delivery of services, IBM may not have adequate contractual cover in situations where contracts are not in place on a continued basis - Unidentified bypass offenders may result in further bypasses
Risk Acceptances	Review the nature of any risk acceptances raised by the business during the audit period and/or any pending approval. Obtain action plans to address identified risks	- Risk acceptances may indicate process weaknesses, consideration should be given whether issues identified on one contract have been reviewed for other occurrences (within a portfolio) - Review the actions taken to meet deadlines, and reasons for committed deadlines to have been missed.
Applications and IT	Review the application listing to determine if there are any unique systems outside of BAU support systems (pricing, billing, revenue reporting etc.). Review should also include application access.	- Relevant systems may not have the required ASCA certifications - Continued Business Need check not performed, and employees may move to other roles leading to SOD conflicts - Process or control weaknesses should be considered in the context of application controls, in particular reconciliations between systems
Management Self Assessments	Review the basis for MSA ratings. - Review MSA reports and understand the justification for the rating - Identify instances where ratings have changed and assess whether information supports the changed rating.	- Incorrect assessments may limit management's visibility of issues on a contract, portfolio, or business area. - Information supporting MSA ratings may indicate process weaknesses requiring further review

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Possible Concerns
Business Controls/Peer and Other reviews	Review other reports / reviews recently performed - Obtain reports by other Internal Audit teams (whereby a CRMfS audit follows an ITD audit as an example) - Obtain reports by business controls, peer reviews, and reviews by the customer where relevant	Indications may be given on areas for further testing by Audit or follow on testing. Consideration should be given to the timing of reviews (focusing testing after the review), and assessment of whether management fully addressed the risk through their actions.
Documents of Understanding (DOU)	Determine if DOU's exist between the auditee and other IBM organizations Review DOU's that directly support service delivery and contractual commitments.	Missing DOU's or incomplete DOU's identified during solution design and delivery may impact the contracts' ability to meet service levels. In particular, for contracts requiring resources and services from other lines of business, assess whether the engagement team have put DOU's in place to meet contracted commitments (equivalent to back-to-back agreements with third parties).

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PLANNING: Reference Material

Please see the sections below for a listing of all applicable Corporate Instructions, Policy Memos, and Procedures Manuals

Solution Design and Delivery Process

All information related to the Solution Design and Delivery (SDD) process, tools, status deployment, and Geography Specific information are available in the following intranet sites and should be consulted prior to the engagement.

WW Solution Design and Delivery Tools

Review file maintained by IBM Consulting CRM Global Process Focal providing a high-level list of tools used in the process. Note it is not intended to be a trusted source and used only as a guide.

For Geography Specific information, contact the SDD process leaders at the geo

level. <https://w3.ibm.com/w3publisher/ibm-consulting-crm-process/others/foundational/tools-templates>

Other tools used as part of the SDD process (during the audit access may need to be granted)

- BI Central: Used to request report access and for generating and formatting claim time reports.
<https://bi-central-portal.us1a.cirrus.ibm.com/#/home>
- Bidlog: Used for pricing and assuring T&M RFS.
- BOSS (Billing and Opportunity Services System) - Used for lab services contracts and time recording)
- CFT/s: Customer Fulfilment tool for services
- Conga: to support Cybersecurity Services (CSS) deals
- Customer claims database
- DCA (Deal Compliance Approval Tool): Various deal approvals and supporting documentation repository
- ECR (IBM Enterprise Contract Repository)
- ESSBASE / COGNOS TM1: Financial consolidation tool used by all contracts.
- EnableMe: pre-contract work approval and registration
- IBM Concessions: concessions reporting and approval
- IBM Engage Support: global Q2C platform
- ILC (Internet Labor Claiming): Accurate accounting of time and allocation of labor cost allowing appropriate recovery.
- GreenSTAR: IBM Consulting's integrated, AI-powered engagement platform (deploying 2025)
- Prophet/Book of Business: Project control database. Used for delivery excellence process. (EMEA IBM CONSULTING)
- QAWB: Quality Assurance Work Bench - repository for QA / risk reports and the associated GSRisk output
- RBK (Release Builder Kit): end-to-end intelligent workflow to help Opportunity Management Team manage the deal process
- SDM GPE: Solution Definition Manager Gross Profit Indicator. Costing and pricing tools
- SRA Tool: Self Risk Assessment tool.
- SSC (Security Solution Community): to support Cybersecurity Services (CSS) deals
- WWGPE: Services pricing tool (principally used in IBM CONSULTING and due to be sunset).

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PLANNING: Reference Material (Continued)

Links to Specific SDD Areas

For information related to the following areas, please refer to the links/intranet sites listed below.

Enterprise Risk Management: <https://w3.ibm.com/w3publisher/enterprise-risk-management>

IBM Consulting SIH Review Managers Wiki: <https://w3.ibm.com/w3publisher/gbs-sih-review-managers-wiki>

Global IBM Consulting Self-Assessment & Risk Consulting - Contract and SOW Review Guidance
<https://w3.ibm.com/services/lighthouse/documents/168016>

Worldwide Records Management

<https://w3.ibm.com/ibm/corporate-documents/worldwide-records-management/worldwide-records-management-plan>

These may be useful however when reviewing CRMfS note that these processes have their own auditable entities and programs.

- Services WW Signings and Backlog Guidance
<https://w3.ibm.com/w3publisher/services-ww-signings-and-backlog/s-bl-guidelines>
- Services WW Signings and Documentation
<https://w3.ibm.com/w3publisher/services-ww-signings-and-backlog/process-documentation>
- Global Procurement Policies & Practices
<https://w3.ibm.com/w3publisher/procurement>

Links to Business Units and Processes

IBM Business units

- IBM Consulting: <https://w3.ibm.com/w3publisher/ibm-consulting>
- IBM Consulting GIC: <https://w3.ibm.com/w3publisher/globally-integrated-capabilities>
- IBM Consulting Essentials: https://w3.ibm.com/w3publisher/gbs_europe/essentials
- TLS: <https://w3.ibm.com/w3publisher/tss-global-offerings>

IBM tools and processes

- Program Management System (PgMS) WW Homepage: (Note: PgMS replaced PMP)
- Worldwide Project Management Method (WWPMT):
<https://w3.ibm.com/project-services/tools/methods>

Relevant Accounting Practices and Instructions

- Accounting Policies and Instructions:
<https://w3.ibm.com/w3publisher/accounting-policy-financial-reporting>
- AI-CTRL 12-014 Accrual and Journal Clip Levels
<https://ibm.ent.box.com/file/724976846286?s=rxakw3jmssn50okgr0l4fjatykn3kbnr>
- Worldwide Chart of Accounts
<https://w3.ibm.com/w3publisher/wwcoa>
- IBM Accounting: <https://w3.ibm.com/w3publisher/ibmaccounting>

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PLANNING: Reference Material (Continued)

Corporate Documents

All Corporate Directives (including Corporate Instructions and Corporate Policies)

<https://w3.ibm.com/ibm/corporate-documents/>

Corporate Instructions

<https://w3.ibm.com/ibm/corporate-documents/corporate-directives/corporate-instructions>

Business Conduct Guidelines

<https://w3.ibm.com/ibm/corporate-documents/business-conduct-guidelines>

Particularly Relevant Corporate Directives and Policies

Legal Corporate Instruction Letters - **All**

Finance Corporate Instruction Letters - **All**

Intellectual Property and Licensing

- [IPL101: Rights to Intellectual Property](#) resulting from work performed by or for IBM under contract
- [IPL106: Receipt and disclosure of confidential information](#)

Environmental Affairs and Product Safety

- [ENV211: Safeness of IBM products](#)

Standards

- IBM Global Chief Data Office: <https://w3.ibm.com/w3publisher/ibm-global-chief-data-office>
- IBM Cybersecurity – PSG: <https://pages.github.ibm.com/ciso-psg/main/>
- IT Security Standard: <https://pages.github.ibm.com/ciso-psg/main/standards/itss.html>
- Security Standard for IT Traffic in Arms Regulation (ITAR) - (ITCS 114)
<https://pages.github.ibm.com/ciso-psg/main/standards/itcs114.html>

Finance

- IBM Finance: <https://w3.ibm.com/w3publisher/finance>
- IBM Finance and Operations: <https://w3.ibm.com/w3publisher/finance-operations>
- IBM Finance IT: <https://w3.ibm.com/w3publisher/finance-it>
- IBM Finance and Operations - Tools & Resources:
<https://w3.ibm.com/w3publisher/finance-it/f-o-tools-and-resources>
- CA&AS: <https://w3.ibm.com/w3publisher/ca-as-public>

Including:

Finance by unit and finance by process.

Finance applications and tools and how to get access.

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PLANNING: Reference Material (Continued)

Data Security and Privacy

European Commission

- Data Protection: http://ec.europa.eu/justice/data-protection/index_en.htm
- Data transfers out of Europe:
https://ec.europa.eu/info/law/law-topic/data-protection/international-dimension-data-protection/rules-international-data-transfers_en
- Standard contractual clauses for data transfers:
https://ec.europa.eu/info/law/law-topic/data-protection/international-dimension-data-protection/standard-contractual-clauses-scc_en

Information Commissioners Office.

Guidance: <https://ico.org.uk/for-organisations/>

European Law resource

- Home: <http://eur-lex.europa.eu/homepage.html?locale=en>
- Example: <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2010:039:0005:0018:EN:PDF>

IBM Standards and Guidelines

Home: <https://pages.github.ibm.com/ciso-psg/main/>

Privacy@IBM

- Home: <https://w3.ibm.com/ibm/privacy/knowledge-base/knowledge-base-a-z>
- GDPR: <https://w3.ibm.com/ibm/privacy/gdpr>

CAAS Audit Programs

- IBM as Controller: <https://ibm.box.com/s/upn7cqitlsmtabsifj2xfayfqkzddkz>
- IBM as Processor: <https://ibm.box.com/s/6wcvy0ir09ny8p9z7hk85loewh7eit7z>

Workforce Management

[IBM Services Workforce Management](https://w3.ibm.com/w3publisher/ibm-consulting-global-facilitation-center/scope-of-work/when-to-engage-gfc)

<https://w3.ibm.com/w3publisher/ibm-consulting-global-facilitation-center/scope-of-work/when-to-engage-gfc>

See separate WFM Organization for Consulting Review 04.15.22 final.pdf attachment.

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PLANNING: Fraud Template

Fraud is any intentional act or omission designed to deceive others and resulting in the victim suffering a loss and/or the perpetrator achieving a gain. Fraud can be categorized into fraudulent financial reporting, misappropriation of assets, and improper or unauthorized expenditures.

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Bribery - Commitment of services to the customer without approvals of price, or non-standard terms and conditions including side letters in order to meet signings and revenue targets.	Separation of Duties (SOD). Sales Transaction Support (Fulfilment) checks. Compliance testing key controls over operations (KCOs). Certification, reviews, and approvals of the deals.	Verify that price was complete and approved prior to communication to the customer. Review the contract for non-standard terms and conditions and determine if the appropriate approvals have been obtained. Ask if there are any potential side letters. Ensure that any concessions have been fully approved via the claims process.
Fictitious Revenue - Contracts are manually altered, agreements are backdated, or the customer's signature is forged in order to meet signings and revenue targets.	Sales Transaction Support (Fulfilment) checks. Compliance testing key controls over financial reporting (KCFRs). Deal release or certification reviews/processes. Management reviews of aged debt and held billing.	Manual alteration: Review contracts to identify manual alteration or typed text which looks out of place. Backdated agreements: Select contracts with signature dates on the last few days before quarter-ends. Compare signature to contract stamp date or date specified in document properties. Verify whether signature date is on a weekend or when the person who signed the contract was on vacation. Compare pricing approval date with the signature date.

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PLANNING: Fraud Template (Continued)

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Kickbacks - The fair value of IBM purchases is not determined or paid for services are not delivered in order to transfer funds from IBM to the person committing the fraud Consider level of testing as Procurement is a separate auditable entity.	Procurement involvement in completing fair value assessments. Procurement controls over sole source and customer directed suppliers. Compliance testing KCOs.	Compare a universe of purchase orders (PO) with a universe of supplier invoices. Test invoices which do not have corresponding PO or where the invoice date is prior to the PO date. Review the supplier contract, determine whether procurement has been involved and fair value has been determined. Determine if there is a linkage between the terms in IBM's contract with the supplier and the terms in IBM's contract with the customer. Determine if there is a demonstrable deliverable for the expenditure. For customer directed suppliers, ensure that the customer instruction is fully documented.
Application Access/Separation of Duties (SOD): The use of applications and databases, specifically those containing personal or sensitive data, as a result of unapproved or unnecessary system access, in order to transfer funds to the person committing the fraud or to provide the data such as, personally identifiable banking information, to another party.	Continued Business Need (CBN) reviews are performed for most applications. System access levels associated to roles to maintain SOD.	Review system access lists including the CBNs performed on the applications. Perform tests to validate some of the transactions performed. Establish if accesses are in line with job roles, and that individuals do not have accesses to multiple ids to circumvent SOD controls.
Financial Recording - Improper signings, revenue recognition or cost recording in order to meet signings, revenue and GP targets, e.g., (i) recording revenue for OEM gross without proper accounting review (ii) recording of revenue for hardware or software dependent on undelivered service (iii) unsubstantiated accruals or premature revenue recording or signings (iv) cost deferred without review and approvals (v) recording of signings without sufficient customer commitment.	Accounting reviews. Management reviews. Compliance testing KCFRs.	Review revenue accruals, signings, contract with OEM content, cost deferrals, contracts which combine hardware and software with services, and revenue reversals after quarter-end. Ensure that information sent to Accounting for review is accurate and enables Accounting to make a fully informed judgement.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

PLANNING: Fraud Template (Continued)

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Theft of assets (such as PCs, servers, or software) as a result of the absence of or insufficient asset reconciliations. Consider level of testing as Asset Management is a separate auditable entity.	Book to floor (B2F) and floor to book (F2B) reconciliations. Movement between locations tracking.	For contracts where IBM own assets or is responsible for the tracking of customer assets, review the asset reconciliations. Select sample of assets and perform reconciliations during the review.

Note: The probability and impact will vary depending on the organization/business unit that is being audited.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Objective:

Are solutions designed in accordance with customer requirements captured during opportunity management, properly reviewed and approved by management, and in compliance with IBM guidelines and instructions.

Risk/Exposure:

The design phase of a solution is crucial to the success of a project. A complete and realistic evaluation of the risks and costs (both internal and external, e.g., suppliers) associated with the proposal is extremely important, and a key input is a full understanding of customers' expectations. Also, a thorough evaluation and approval of all pricing elements and non-standard terms and conditions is necessary to prevent legal issues and safeguard IBM's profitability.

For portfolio audits obtain following information for each selected sample as part of testing the design elements (process review / RFI phase):

- A presentation for each of the selected contracts. The presentation should cover: the contract structure, its management system and financial management, status of any transition and transformation programs, key contract changes, requests for service, and the status of any identified issues and associated action plans. It should include an explanation of the provided inception-to-date financial analysis with the explanation for any significant variances from the original Pricing Case.
- A soft copy of the contract and any significant amendments, including all schedules addendums, annexes, signature page, etc.
- A report of all Contract Changes (CC), Project Change Requests (PCR), and Request for Services (RFS) completed or in progress during the audit period, including: CC/PCR/RFS ID Number, Contract Number, Customer Name, Customer Number, Title, Reference Number, Status, Date Contract Signed, Effective Date, Type of Service, BTT (or equivalent assessment), Brief Description of the Change/Request, Value and Planned GP %.
- If the contract is less than two years old - A copy of the original approval deck and Risk/Quality Assurance Reviews/Assessments performed on the contract prior to signature.
- If the contract is less than two years old - The final approved pricing case for the contract and all associated approvals. Note: this is the pricing case at contract signature.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Primary Test: Solution Design Mandatories (SD001.P)

Objective:

To ensure that the opportunity has been progressed using the mandatory IBM processes and procedures.

Approach:

Assess the complexity of the opportunity and determine the mandatory activities required. For the selected opportunities verify the following:

- Was the complexity calculated and completed correctly, including any subsequent adjustments for changes in the final stages of contract negotiation. Complexity and approval requirements can be determined differently depending on the line of business e.g., PATC (Primary Approval Table Calculator) or BTT (Business Transaction Type).
- Where appropriate was centralized/specialized solutioning function engaged or exemption obtained
- All internal approvals were in place before the proposed solution was submitted to the customer.
- Establish if there were any conditions to approvals and if so, were those conditions met.
- Was the customer's credit worthiness assessed.
- Does the customer and the solution meet US Export Regulation requirements. Confirm that written procedures and controls are in place that require all original solution proposals or substantive changes to proposals to be evaluated for export regulation requirements in accordance with the provisions of USERP Section 6 Part 1,3 or that solution qualified for an exemption under USERP Section 6, Part 1.5. Also check that records demonstrating compliance with IBM's export regulation compliance requirements are being retained in an accessible manner for the required time period. (USERP Section 2, Part 2.6.1 and Section 6, Part 1.6) These records primarily include but may not be limited to (a) completed IBM Client Services Evaluation Guides (GSEGs), (b) CSEG exemption eligibility documentation, (c) records documenting resolution of 'yes' responses on CSEGs and (d) sending country 'written confirmations' to offshore global delivery centers (when engaging resources during Services Solution Delivery).
- Was a denied parties listing check carried out.
- Is there evidence that the customer requirements were clearly and comprehensively defined, understood, and agreed. Is there evidence that a detailed solution that addressed the requirements was documented within the contract and supporting statements of work.
- Is there evidence of proposal and other supporting plans in place (opportunity plan, staff, and resource plans).
- Was the pricing case created, approved, and released by delegated and authorized persons using approved pricing methods and tools.
- Was the solution correctly priced (refer to test SD002.P for more details) based on a complete understanding of the solution components. Factors such as: transition and transformation were well understood and included in the price, non-standards clauses were priced for such as trade-in, buyback, demonstration, or trial period offering, resource efforts and mixes were accurate, contingencies, currency fluctuations, etc.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Primary Test: Solution Design Mandatories (SD001.P)

(Continued)

- If a deal is cross brand, was pricing approval evident for all lines of business involved. How was it ensured that all approvals are in place prior to price release.
- If a deal is cross brand were pricing approvals based on a consistent understanding of the solution.
- Was there evidence of a pricing commitment being made to the customer prior to approval. Was a final fully approved price release letter (or equivalent) in place before contracting.
- Where required, were other reviews and assessments completed and approved at the right time by the correct people. Different lines of business require different reviews, some examples are self-assessment, peer assessments, technical delivery assurance reviews (TDA), integrated technical reviews (ITR), delivery excellence reviews, global facilitation center (for cross-border / international) or other technical, quality and risk management, or other solution reviews.
- Where appropriate other organizations have reviewed the solution and the contract such as C&N, Legal, Accounting, etc.
- Terms and conditions were properly selected and approved and if non-standard terms and conditions were used, they had the appropriate review and approval.
- Is there evidence of any subsequent amendments to terms and conditions and were they identified and approved.
- A preliminary customer inventory was available where required. (The contract should include a clause allowing for financial adjustment when the final inventory is established), and a plan in place to confirm the inventory.
- If third parties were used, were agreements designed to meet both IBM and customer requirements. Were agreements of services and other deliverables back-to-back and were resources secured as required.
- Where OEM products are part of the solution, ensure that Product Safety Review Board (PSRB) approval of each product has been confirmed.
- If applicable Open Source Software or Environmental Affairs and Compliance reviews were complete. Audit program 47-10-00 will cover these areas in general and Audit program 44-00-00 (section 12)

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Primary Test: Pricing (SD002.P)

Objective:

Determine if the solution was correctly priced and approved and if all contractual obligations of IBM were adequately reviewed and properly authorized.

Approach:

Assess if mandatory pricing activities were performed. These may differ based on the line of business, profitability such as GP or PTI, or complexity. For the selected opportunities verify the following:

Mandatory Checks

- The firm price offered to the customer was approved within the delegated limits of the pricer and the organization.
- Any pricing that was not firm (ROM / Indicative, etc.) that was presented to the customer had the required wording characteristics included that limited IBMs commitment to that price or allowing increase / decrease in the price and that it also had the correct approvals.
- All elements of pricing were approved prior to communication to the customer.
- The pricing includes the recommended contingencies for risks.
- The pricing includes all cost considerations including costs of subcontractors, vendors, penalties, termination charges, due diligence activity cost, quantified risks, baselines from joint verification / inventory assessment and costs to perform those, hot skills, inflation (COLA, Cost of living adjustment), foreign exchange, data security and privacy and resource mix (onshore/offshore or banding).
- The pricing is based on approved labor rates.
- The price was developed in accordance with published methodology and using approved pricing tools at the time.
- The complexity of support processes was included in the price. (e.g., costs associated with billing requirements, additional support from other IBM support functions e.g., C&N, Procurement)
- Transfers of staff were included in costing, including an evaluation of differences in benefits and compensation for customer staff transferring to IBM.
- The pricing case includes the correct level of revenue apportionment (indirect costs).
- Where IBM Global Financing is part of the offering to the customer, are all elements of the price consistently documented, and financing based upon the final approved price.
- Late changes to the proposal price were appropriately reviewed, and re-approved where necessary and documented (e.g., changes in scope, total contract value, T&C's, concessions, executive decision, or revenue credit activities).
- A formal price release letter (PRL) from an authorized source, global price estimator (GPE) or equivalent was issued.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Primary Test: Pricing (SD002.P)

(Continued)

- Pricing subject matter experts were properly informed and involved in deals where necessary.
- Where additional business inputs were needed, reviews with appropriate participants occurred (e.g., Global Delivery, Global Facilitation Centre or Procurement).
- If tools were used to generate the required pricing approvals, establish if they were reviewed in line with published and delegations in a timely manner.
- If deals were across multiple lines of business or service lines, understand if there are any additional requirements or impacts to pricing. Determine how elements were combined to deliver a complete and accurate final price case where revenue could be correctly apportioned.
- Cost case assumptions on transition and transformation, including planned cost deferrals were in line with accounting guidelines.

Quality of Pricing

- The pricing case took into account the current contractual position. If the proposal related to an existing contract, then there may have been financial issues associated with the current contract which would need to have been addressed in the renegotiated contract, in terms of pricing and the revised contract terms. Where the contract is an extension of a previous contract, check that issues identified in previous risk management or quality assurance type reviews have been appropriately addressed in the pricing of the extension.
- The contract price was documented within a complete and clear billing plan and transferred to Sales Transaction Support (fulfilment) accurately and timely to enable subsequent billing to occur as required.
- For contracts that include growth in business volumes where pricing assumptions are based upon accurate customer input on future volumes. Verify what actions have been taken to provide an independent and objective assessment of the validity of those assumptions.
- The pricing of the solution has been reviewed for completeness to identify whether there are any gaps from the contracted solution e.g., all customer business units in scope are part of costing.
- Costs were realistic and addressed risks identified pre-signature. Complexity of the solution was identified, and the impacts on time and resources quantified e.g., on an SO contract, transition of complex infrastructure/multiple countries/parallel projects/transition direct from customer to IBM Global Delivery Centers; management of IT services to very high SLAs (e.g., 99.9%).
- Where pricing include third party costs for services and/or OEM equipment, validate the actions taken to ensure the accuracy of those costs. Have procurement obtained commitments (in the form of proposals) from suppliers that support the pricing to IBM's customer? Where negotiations extend beyond the initial validity period, are proposals from suppliers revalidated? For higher complexity and large contracts, and contracts with high percentage of third party spend, supplier quotations and terms should support both pricing and IBM customer terms and conditions.
- Where IBM has estimated third-party costs, assess the involvement of Procurement in the review of supplier costs, and that appropriate comparisons have been performed based upon required skills and number of resources.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Primary Test: Contract Design and Agreement (SD003.P)

Objective:

To ensure that contract terms can be met by IBM and that they provide certainty to both parties and protect IBM from risks. To that end, the contract should ensure that customer requirements were clearly and comprehensively defined and documented in the contract, and that the contract is consistent with all approvals and key assumptions in solution design.

Approach:

The contract should be reviewed for non-standard clauses. If such clauses are identified consideration should be given to the impact on the other CRM activities, such as revenue recognition, financial management, approval requirements, etc.

Examples of clauses to look out for include:

- Unrestricted sweeper clauses which can be financially detrimental to IBM. A sweeper clause is any clause that commits IBM to unlimited, unquantified, or undefined services and/or liabilities. Indicators of a sweeper clauses include ongoing and increasing cost overruns, increased scope, and services without associated revenue and missed service levels.
- Lack of a punitive termination clause or re-baselining of the contract to circumvent termination clauses.
- Unrealistic SLAs or targets. Consideration must be given to the type of service and industry (e.g., high levels of server availability may be expected where IBM support customer systems, however, look for SLA where the requirement exceeds typical industry standards or are set at 100%).
- SLAs which have a dependency on a third party outside IBM's control (e.g., end-to-end SLAs where IBM does not have responsibility for all of the service).
- Ambiguous contract language.
- Review instances of ongoing and protracted negotiation (e.g., for contract changes).
- Retrospective bench-marking clauses.
- Lack of clear dispute resolution path or a path that is detrimental to IBM.
- Access to IBM's audit and investigation reports - also review instances where IBM is required to be involved in customer audit reviews, and impact on costs and actions required to address identified issues.
- Minimum revenue guarantees. In particular, are there any terms that restrict the achievement of minimum revenue levels, impact IBM's ability to cover its fixed costs? Can customer "cherry pick" services and is that addressed in the pricing and risk associated with the contract.
- Terms that impact IBM's Intellectual Property rights over solutions.
- Warranty terms that are non-standard or onerous.
- Terms that are difficult to monitor and control where IBM would have liability, such as IBM being responsible for software ownership and licensing for a customer.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Primary Test: Contract Design and Agreement (SD003.P)

(Continued)

In addition, the contract should be reviewed to ensure that:

- The contract included a defined statement of work that included clear and measurable acceptance criteria where applicable. Deliverables and services are clearly defined with clear performance indicators and timescales where appropriate.
- Where a high-level design is documented, review the timescales and actions required to agree upon a detailed solution design. Ensure the responsibilities for completion are clearly defined. Ensure that if design is not fully completed, then the ability to charge for additional work identified during 'full design' is included in the contract.
- When only a preliminary customer inventory was available, the contract included a clause allowing for financial adjustment when the final inventory is established and timing for determining final inventories. Where customer assumptions are provided, validate that these were from customer systems and recent.
- For incremental business, the impact on the scope, deliverables, billing, and service levels of the base contract has been clearly identified.
- A review has been performed of the contract design to ensure that the solution design is accurately and consistently documented into the contract.
- IBM resources are available to meet the customer requirement. Where resource commitments are required from other lines of business (e.g., IBM Consulting resources to deliver an outsourcing contract or the use of global resourcing) what actions have been taken to confirm the availability of resources at the right time with the appropriate skills. Have DOUs been established that secure resource at the right skill level to deliver solutions. Also, that resource mix is agreed and correct and doesn't pose as a conflict to client requirements.
- If IBM resources are not available, ensure that sub-contracted resource is identified and contracted through IBM Procurement. Ensure that deliverables are at a minimum matching the customer contract and that the use of subcontractors is permitted within the customer contract.
- Whether there are any onerous terms associated with customer disputes e.g., ability for the customer to withhold payments.
- Is there evidence of contract changes to address original contract or solution design weaknesses.
- That the scope of work performed by Global Resource as part of a Cross Border process is detailed in the contract, including any limitations. Can signed contract change documentation be produced?

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Primary Test: Contract Design and Agreement (SD003.P)

(Continued)

Contract approvals:

- Was the credit status re-evaluated by IBM Global Finance before proposal was issued to the customer.
- Have all conditions in the approval of the contract (e.g., by Pricing, Legal, Accounting, Risk Management or Quality Assurance type equivalents) been met? If not, determine why not.
- Are approvals ambiguous or implicit, e.g., by copy of an e-mail / attendance at a review meeting rather than explicitly and clearly provided.
- Check for any unapproved conditions inconsistent with the contract (e.g., side letters; written or verbal).
- Were the contracts signed and all approvals obtained before commencement of supply of service? If not, was there an approved letter of intent (LOI) or authorization to proceed (ATP) in place? Check also SE009.P testing for work at risk or pre-contract work.
- Was there evidence that contract signatures had been back dated or that the contract had been changed by the customer before signature?
- Have required opportunity learning reviews (formerly Win Loss reviews) have been conducted and documented according to the appropriate clip-level? Check that procedures were in use to disseminate lessons learned.

Data privacy considerations:

- If personal information or sensitive personal information will be collected, transferred, or processed in another country, does the contract include the required data privacy terms, and describe the data protection responsibilities.

Business partner / third party considerations:

- When the contract for services is between IBM and a business partner or other third party (like systems integrator, prime contractor, or legal entity other than the end-user customer) was the applicable contract template / terms used

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Secondary Test: Risk Management - Design (SD004.S)

Objective:

To ensure that the required risk management activities are undertaken during the engagement phase, during contract preparation and pricing, and ongoing through delivery. (see also Contract Solution Delivery SE004.P). Reviews can be across the whole contract or more focused on technical, or financial/accounting issues.

In different lines of business (LOB), risk management activities are undertaken by various teams including Risk Management, Finance, and Business Controls. The process of independent review forms an integral part of the Solution Design and Delivery process and is intended to help ensure profitable, high-quality service solutions. The reviews are performed initially during qualification, proposal development and then throughout project delivery. Their objective is the recognition, containment, and / or mitigation of any "risks" that would jeopardize the success of contract. Specifically for services, quality assurance is a critical factor for success. Particular focus on larger contracts would be in the pre-contract signing, transition to business as usual and ongoing transformation stages.

Approach:

Risk management has been developed to minimize the common, and often repeated risks, to IBM that are inherent in customized solutions and services. Risk management activities therefore assess whether contract / IBM objectives can be met based upon the solution that has been designed, whether risks have been identified and what actions were put in place to effectively manage those risks.

For the relevant sample of contracts:

- Determine whether the mandatory risk management and quality assurance type processes and reviews were sufficient to identify and address the risk in the contract.
- Identify the minimum requirements for mandatory reviews and evaluate that those were met. Identify that the right reviews occurred, at the right time by suitably qualified individuals. Reviews depend on the type of solution, the line of business and the contract value and examples can include the following: Preliminary opportunity owner self-assessment, peer partner review, technical delivery assessments (TDA), contract baseline review (CBR) , integrated technical review (ITR), proposal baseline assessment (PBA) or other technical, commercial or business reviews required.
- Were reviews sufficiently comprehensive and involved the appropriate subject matter expertise based upon the indicators of risk in the solution (e.g., were non-standard multi-currency billing terms reviewed by Finance and Treasury where required).
- Evidence of effective risk management activities occurring throughout solution design and delivery processes: Risk management logs (e.g., IBM Program Work Centre, IPWC) are comprehensive and kept current, risk management plans exist that include ownership, action plans, priority, due dates and are regularly reviewed with the correct participants and outcomes agreed.
- For cross line of business contracts, has the relevant subject matter expert / appropriate lines of business reviewed and documented their approval of the technical solution, pricing and schedules. Have adequate actions been put in place in the internal governance processes to enable ongoing management of the contract.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Secondary Test: Risk Management - Design (SD004.S)

(Continued)

- Is IBM able to deliver the solution that has been originally defined in the contract? Are any expectations unclear, unrealistic, or undeliverable (technically, or for other reasons such as resource and skills availability).
- Were there significant changes to the contract in the final stages of negotiation that significantly impacted the feasibility or the solution and have actions to mitigate those risks been established?
- Where changes to the solution have been made since the original mandatory reviews (e.g., technical design assessment), was the impact raised and quantified in final approval decisions? Were reviews re-performed if necessary.
- Where multiple indicators of risk exist in the solution, are additional reviews performed to quantify the impact. Examples of risk indicators at design stage include: unsecured resources, key resources unconfirmed, multiple iterations of the solution, pricing that is materially inconsistent with sector average pricing, multiple high-risk ratings for mandatory reviews (Peer Partner reviews, TDA, ITR, etc.).
- Were risk management activities performed in a timely manner, with sufficiently skilled personnel. In addition, verify that recommendations and actions are tracked through to closure. Were required functional approvals, and signoffs in place (e.g., legal, contract and negotiations, procurement, HR, Accounting and Finance). Were reviews performed at appropriate times?

Review the quality of the risk management reviews performed:

- Did the reviewer come to the same conclusion relating to the impact of issues as the auditor, and where differences exist, which aspect of the review process was not operating correctly. Has the analysis completed by the reviewer been performed to the required standards?
- Do audit agree with QA/risk management ratings. As an example, QA rate reviews from A (highest) to D (lowest) based upon defined criteria, validate the rating of the contract against those criteria, and confirm whether the rating was correct.
- Were due dates agreed and met to close recommendations from risk management activities or did due dates slip. If recommendations are not closed identify why not.

Note: The auditor should consider if findings in tests performed under Solution Delivery were caused by failure to effectively complete risk management / QA type reviews and the required solution design mandates.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Services Solution Design

Primary Test: Contract Change and Request for Service (RFS) (SD005.P)

Objective:

Ensure that adequate, effective, and timely contract change management controls were in place. This should include service requests, in-scope and out-of-scope contract changes, and terms and conditions changes.

A contract change (CC) is typically a change to the base contract affecting terms and conditions, price, or delivery. Requests for service (RFS) are within the scope of the contract and may represent additional revenue or be counted as part of a committed revenue spend. Where requests for services are outside the scope of the base contract, a contract change will also be required. CCs and RFSs are equally subject to solution design and delivery processes. Review the requirements in the contract design and agreement process for CCs and RFSs as these agreements are subject to the same mandatory requirements dependent on clip levels and/or complexity.

Approach:

Obtain the universe of CCs and RFSs during the audit period. Identify high risk samples based upon value, financial performance, sign date (close to quarter-end), complexity, penalties or settlements and select a sample of those to identify if change and request processes were followed:

- Were all CCs and RFSs supported by the documented and line of business solution design techniques, approvals, and appropriate contractual documentation? (see also SD001 and SD002)
- Where third parties support IBM's turnaround targets, are back-to-back terms in place to enable IBM to meet its targets and recover any penalties from the third party in line with the customer contract?
- Review the change management process to manage scope changes to RFSs and assess whether all the changes were correctly identified and priced.
- Where variances occur, was a root cause or common issue(s) identified and addressed in the solution design of future RFSs?
- Where appropriate, confirm that RFSs are rolled up into CC on a regular basis. How are roll-up requirements determined by the business and implemented by the contract team.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Objective:

Are solutions delivered in accordance with commitments? Are internal and external resources monitored and utilized effectively?

Risk/Exposure:

Because of the complexity of most services' solution delivery projects, the execution phase requires a particular focus on IBM performance, at any point in time, versus the contract commitments and the customer expectations. The lack of project follow-up and of monitoring for subcontracted services, and the failure to effectively use or re-allocate resources could jeopardize IBM's ability to fulfill the contractual commitments and result in cost overruns, negatively impacting profit.

For portfolio audits obtain following information for each selected sample as part of testing the delivery elements (process review / RFI phase):

- A presentation for each of the selected contracts. The presentation should cover: the contract structure, its management system and financial management, status of any transition and transformation programs, key contract changes, requests for service, and the status of any identified issues and associated action plans. It should include an explanation of the provided inception-to-date financial analysis with the explanation for any significant variances from the original Pricing Case.
- A summary list of all contract deliverables with original target dates and status.
- A copy of all completed and signed milestones/deliverables.
- An inception-to-date financial analysis, including a full explanation for any significant variances from the original Pricing Case.
- Asset control procedures applicable to the contract, including hardware and software inventory procedures, include vendor managed assets.
- A report of the monthly balance sheet (contract balances), including reconciliations.
- A copy of all Project Management Reviews and other internal reviews performed on the contract since contract signature (or in the last 2 years if the contract was signed more than 2 years ago).
- A copy of the last Book-to-Bill reconciliation report performed on the contract.
- Result of customer satisfaction surveys conducted during the audit period and related actions plans for the selected contracts.
- Copies of executive management customer satisfaction reviews performed since contract signature.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Solution Startup and Implementation (SE001.P)

Objective:

Determine that a transition and transformation (T&T) plan, including employee transfer (if required) was documented and executed to ensure all requirements were met to implement delivery strategies.

Note: Transition and transformation (T&T) plans are applicable for contracts where the customer has outsourced their IT, business, or application services. Transition is the transfer of services from the customer / other vendors to IBM, and a requirement for billing for services. Transformation is typically a key element of reducing costs on many contracts to achieve contractual commitments to reduce price in future periods, e.g., server consolidation, process re-engineering.

Approach:

Assess the complexity of the opportunity in addition to determining the mandatory activities required according to the business line requirements such as Business Transaction Type (BTT) or Risk Complexity Assessment (RCA). In particular, establish the approach to transition:

- Was T&T activity from the customer to IBM or was a previous third-party provider to the customer involved?
- Was T&T from the customer to IBM "in country", or to an IBM Global Delivery Centre (GDC). Where GDC's are part of the T&T plan, was there a period of parallel running? Was this in accordance with original plans or delayed?
- Were processes operated by the customer sufficiently understood and agreed upon to enable effective T&T?

Note: The auditor should consider if findings in other tests performed under Solution Delivery were caused by failure to complete transition phase activities.

Review the transition and transformation (T&T) phase including:

- Was a T&T plan prepared and executed? Were the risks in the proposal and pre-contract solution assurance type reviews (e.g., risk management / QA activities) addressed in the T&T plan?
- If issues were identified in T&T programs on other contracts (within the IMT line of business / sector), were lessons learnt factored into new T&T plans. How are issues formally captured, and incorporated into future risk management activities?
- Were any issues recurrent and already known through lessons learned reviews in other contracts? If there were recurring issues, was root cause analysis performed effectively.
- Was transition and transformation (T&T) completed in accordance with contract requirements, e.g., expected timelines or delays, in line with reduced pricing and cost assumptions / other contract objectives?
- Was post-contract signing due diligence performed and were any issues identified promptly resolved. Did the due diligence highlight issues not identified during contract design? If so, why were issues not identified and what was the root cause of those issues.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Solution Startup and Implementation (SE001.P)

(Continued)

- Were T&T activities well controlled. Is there evidence of delays and were any delays significant or repeated. Were any milestones missed. Were KPIs or other performance measurements in place and were they accurately reported, tracked, and acted upon. If there were misses, what was the root cause of those and were they identified and managed by the contract team?
- Where delays have occurred, assess whether issues relate to design or delivery of the solution.
- Where transition and/or transformation programs are required to reduce IBM's cost of delivery, has analysis of T&T delays been factored into the contract financials and reported for revenue recognition evaluation.
- Were the transition deliverables accepted by the customer, and did they provide adequate basis for continuation of the project. For a sample of deliverables from the T&T phase, assess whether they meet the requirements of the plan, e.g., are key assumptions required to achieve customer commitments validated, are there significant exclusions in the completion of the deliverable?
- Compare planned revenue and cost according to the pricing case to actual revenue and cost. Were there any overruns? If so, understand the root cause (solution design or delivery), understand what was done to fix, and how this was tracked and resolved.
- Check that the resource mix used matches the plan in relation to business-as-usual service delivery and T&T.
- Was T&T adequately reviewed as part of a solution assurance type review (e.g., risk management of quality, or technical assurance reviews)? Was there sufficient solution assurance and review focus placed on the contract early in the T&T life cycle to enable issues identified to be resolved?
- Was labor claiming for transition costs accurate, both in terms of the planned price case and actual labor claiming. Were there any hours not claimed or hours disputed by the customer.
- Were there any large adjustments or write-offs of cost, settlements, claims, disputes, large credit notes, unpaid invoices to determine if any issues exist, not identified in any if the tests above.
- Where there any re-baseline activities, if so, were they appropriate and approved?
- Verify that documented procedures are in place to ensure that sending country 'written confirmations' are included in the workload transition document packages, in accordance with the requirements of USERP Section 6, Part 1.4.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Deliverables (SE002.P)

Objective:

To determine if management checkpoints and controls were implemented for contract deliverables to ensure they were on schedule, within budgeted cost and in accordance with contract requirements.

A deliverable (according to PgMS) is defined as a physical (including electronic) item required to be provided to the Customer by IBM or to IBM by the Customer under the terms of a contract. Deliverables may include hardware, software, or reports. Deliverables do not include services.

Approach:

As part of the process review request a pictorial (or flowchart) representation of the systems used to manage and deliver the contracts, including contract management, delivery activities and system interfaces.

Assess the complexity and determine the total control value (TCV) of the opportunity to establish what mandatory activities are required according to business line. Complexity such as Risk Complexity Assessment (RCA), Business Transaction Type (BTT) can lead to different controls, reviews of requirements. Alternatively, it can be TCV that determines what checkpoint and controls are needed.

Review the contract to understand the deliverables and request the universe of all deliverables and performance against those:

- Were deliverables in the statement of work (SOW) clearly defined?
- Were deliverables defined and agreed in accordance with contract timelines or were they delayed?
- Did approved deliverables meet the standards defined in the contract? Do deliverables contain omissions / scope reduction from the originally defined SoW, if so, what is the impact of the changed scope of the deliverable?
- Were customer acceptance criteria and completion criteria accurately defined in the contract?
- Was evidence retained of customer acceptance of completed deliverables?
- Where required, did deliverables meet documented acceptance tests levels and is there evidence of that the customer has signed off and agreed to performance and results.
- Assess the root cause of instances where deliverables have not been met, is this due to solution design or delivery.
- Is there evidence that the contract team performed root cause analysis where deliverables were not met and implemented effective corrective activities to prevent reoccurrence.
- Were any disputes, concessions, claims, write-offs, aged debt or settlements due to delayed or unclear deliverables?
- If so, are they a one-time occurrence or is there evidence of repeat and ongoing delivery issues.
- Was progress on completion of deliverables effectively reported, tracked, including risks relating to successful completion?

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Deliverables (SE002.P)

(Continued)

- For ongoing or regular deliverables is there evidence that they are reported to and reviewed with the customer as required?
- Has an exit management plan been defined as a contract deliverable that details the responsibilities, and costs for each party in the event of termination of the contract?
- In the case of contracts being terminated, was the contract plan followed and was the contract closure adequately managed? Have all financial closing activities been completed on the contract, including reconciliation of the contract revenue and billing.
- Changes in deliverables or performance issues associated to deliverables may have an impact on revenue recognition. It is important that if issues are identified this should be considered or highlighted to an auditor reviewing revenue recognition and financial management.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Asset Management (SE003.P)

Objective:

To ensure that IBM and customer assets and assets records are managed to enable contracted service levels to be met, adherence to contract terms, and effective interlocks with related delivery processes.

When assessing the extent of testing in this area or determining if this area should be included in the scope of the audit, consider whether the asset management audit program (52-50-00) for the geography would already cover it to prevent duplicate effort.

When considering whether to include asset management testing, you should consider the age of the contract and understand whether basic asset management requirements are being fulfilled. Testing will be more relevant during the early stages of the contract, during transition and transformation or immediately thereafter, to ensure that an accurate baseline has been established, and asset management responsibilities have been picked up correctly from the customer. For older contracts the risk will be much lower as long as reconciliations of both hardware and software are performed regularly to ensure that contractual responsibilities are being fulfilled, and that billing to the customer is accurate.

Approach:

Review the contract(s) and associated schedules to determine if contract specific requirements for asset management were being achieved in line with agreements. This should include IBM or customer owned or leased assets.

For contracts where assets are in the process of, or have been, recently transitioned:

- Review the process and plan established to confirm the assets under scope of transition. What was the method for establishing an accurate baseline.
- If joint verification was required to establish baselines, did that occur and was it completed when it was contractually required to be.
- Is there a mechanism to be able to adjust billing accordingly when baseline assumptions are adjusted.
- Where assets have been transferred to IBM and the transition plan has been completed, validate whether initial assumptions in the contract relating to asset volumes were correct, and whether the actual volumes materially impact pricing or profitability assumptions. If so, were subsequent action plans effective to mitigate risks associated to prevent financial loss?
- Are contractual obligations clearly defined and agreed in relation to asset management.
- Was the asset management function clearly informed of scope of responsibilities and assets. Were they engaged with, and provided with, valid information in a timely approach.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Asset Management (SE003.P)

(Continued)

For contracts where asset transition has been completed over a year ago:

- Establish at a high level that standard IBM asset management processes are in place.
- Validate that contractual obligations are clearly defined and agreed to.
- Determine that steady state contractual obligations are being met.

It is important to recognize that the line between one audit program and another is not strictly defined, and testing should depend upon the unique assessment of the risks associated to contract specifics rather than Asset Management process itself. It is important to recognize that when determining the scope and depth of testing, Reconciliations such as book to floor and floor to book should not be performed under a CRM for Services audit.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Contract Change and Request for Service (RFS) (SE004.P)

Objective:

Ensure that adequate, effective, and timely contract change management controls were in place. This should include service requests, in-scope and out-of-scope contract changes, and terms and conditions changes.

A contract change (CC) is typically a change to the base contract affecting terms and conditions, price, or delivery. Requests for service (RFS) are within the scope of the contract and may represent additional revenue or be counted as part of a committed revenue spend. Where requests for services are outside the scope of the base contract, a contract change will also be required. CCs and RFSs are equally subject to solution design and delivery processes. Review the requirements in the contract design and agreement process for CCs and RFSs as these agreements are subject to the same mandatory requirements dependent on clip levels and/or complexity.

Approach:

Obtain the universe of CCs and RFSs during the audit period. Identify high risk samples based upon value, financial performance, sign date (close to quarter-end), complexity, penalties or settlements and select a sample of those to identify if change and request processes were followed:

- Review whether RFSs impact service penalties, and that controls are in place to manage / minimize penalties.
- Have turnaround targets for CCs and RFSs been met? If not, is that identified with effective corrective action plans in place to address?
- Review the process to determine whether an RFS is chargeable or not. What controls are in place to assess whether an RFS was correctly treated? Ensure concessions are not being hidden by counting RFS work as in-scope.
- Assess whether any work was performed before the RFS was finalized and approved by both IBM and the customer. Are there effective processes in place to identify and track pre-contract work or work at risk. Are there effective processes operating to manage and monitor any pre-approved pre-contract work to ensure it does not exceed business limits. For instance, where pre-contract work / work at risk is an integral part of the process to manage RFSs, check for bypasses and undetected bypasses of these processes to ensure no hours were worked without customer commitment and without additional billing where applicable. (see also SE009)
- Review a sample of deliverables from RFSs and assess whether work performed met the requirements, or whether additional work was performed outside of the scope of the RFS.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Contract Change and Request for Service (RFS) (SE004.P)

(Continued)

- Review the change management process to manage scope changes to RFSs and assess whether all the changes were correctly charged.
- For contracts with significant RFS volumes, has a management system been implemented to monitor and track RFSs for cost and timescales.
- Where variances occur, was a root cause or common issue(s) identified and addressed in the solution delivery of future RFSs?
- Verify that changes to delivery processes required to manage CC and RFSs are incorporated into the management system.
- Confirm that monitoring of RFSs is operating to identify operational delays and cost overruns.
- Assess whether the processes in place to monitor for potential RFS cost overruns are visible prior to limits on RFS being reached (e.g., monitoring to highlight when 80% of the RFS costs are spent) and the processes in place to mitigate against unauthorized over-spend.
- Confirm that baselines impacted by RFSs are correctly priced and adjusted using the CC process. For example, reduced resource charges (RRCs) may be an inappropriate mechanism for adjusting charges following an RFS which decommissioned a large number of servers. This can be particularly significant on a contract which includes bench-marking clauses. Look for RFSs which support significant expansion or contraction of the customer's business.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Procurement and Accounts Payable (SE005.P)

Objective:

To determine whether the supplier to IBM is providing services in accordance with IBM's end customer requirements, and that payments for services are correctly supported.

Note: Detailed testing of the Procurement and Accounts Payable (AP) processes would occur 44-00-00 and 18-10-00 audits. For CRMfS audits, testing should generally be limited to the unique contract risks and deliverables rather than the Procurement and AP processes themselves. Careful consideration should be made when determining what (if any) testing is made and the depth of testing made where there is the potential to duplicate audit effort.

Approach:

Review the solution design for the contract identifying those elements to be delivered by a third party.

- Ensure that the contract team has provided adequate and timely information to organizations responsible for procuring services.
- Validate the involvement of the contract team or line of business in the selection of third-party suppliers, ensuring that adequate understanding of business requirements was in place.
- Review the contract with the suppliers and identify incomplete back-to-back agreements that expose IBM to potential risks such as service level penalties or data security requirements.
- Review actions taken to manage and mitigate the risk of missed service levels in these instances.
- In case agreements are not back-to-back, establish if the contract team have identified this and have implemented appropriate and effective mitigating action plans.
- Establish if there is an effective governance system in place with regular reporting and monitoring of services and other deliverables supplied by third parties.
- Do contractual agreements with third parties detail a governance system and escalation path so that right and responsibilities are well understood and adhered to.
- Where OEM hardware was included in the solution, review whether it was assessed according to the requirements of Corporate Instructions ENV123 and ENV124 for compliance with hardware, environmental, and chemical requirements.
- For a sample of payments to suppliers, ensure that there is evidence of the delivery of services in accordance with the contract requirements, and the review of that delivery by relevant IBM staff. Ensure that the payments are in accordance with the contract with the supplier. For services contracts, assess the adequacy of review by management of services provided. Consider unique contractual risks or unusual payments rather than general AP process.
- In instances where significant bypass activity has been detected on the contract or in the business, and a review of purchases and payments indicates potential unidentified bypasses, review the effectiveness of the management system to monitor procurement activities.
- Where purchases are made on behalf of the customer, assess the adequacy of controls over this activity, and billing to the customer for products / services procured. Ensure that contracted uplifts were correctly applied.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: -2. Services Solution Delivery

Primary Test: Procurement and Accounts Payable (SE005.P)

(Continued)

Subcontractor Management Risks

- Where subcontractors have been used, review the actions taken to assess skills for the role and the availability of IBM staff initially.
- Review continued and ongoing use of subcontractor resource over IBM staff and look at the adequacy of resource planning processes.
- Assess whether an adequate assessment of skills was performed to confirm the availability of subcontractors before committing to a customer project.
- Assess whether there are any specific customer requirements in terms of staffing and establish how that is achieved with subcontractors. For example, levels of background vetting and security clearance.
- Check whether there are specific customer contractual requirements in relation to data security and privacy and establish that this is reflected and operational in third party agreements.
- Check for processes to validate subcontractor invoices for correct labor claiming and other expenses including review and approval by project managers.
- If subcontractors are used, establish that their use is permitted in the customer contractual agreement.
- Look for customer directed sub-contractors' usage without clear evidence of direction from customer.
- Check that standard processes to establish that subcontractor services were validated prior to payment were operation in the contract or business line.
- Where subcontractor services directly impact achievement of service level agreements, confirm that back-to-back agreements, including associated penalties, are in place.
- Where subcontractor services are used and / or terminated, establish if there are effective controls over physical assets, system accesses and intellectual property (IP) so that IBM is protected from loss of equipment, un-authorized access to systems and information and protected against the loss of knowledge and ownership of IP.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Contract Governance and Management System (SE006.P)

Objective:

To ensure that the contract governance processes manage risks in the ongoing delivery of the contract.

Approach:

Consider the complexity and value of the solution and assess the adequacy of external (customer) governance processes and internal (IBM and suppliers) governance processes. Review the following:

- Were contractually required meetings established.
- Are requirements / inputs to those meetings agreed and documented. Is it agreed who needs to supply what information by when to whom.
- For the various levels of contract governance, are escalation criteria defined in the contract to ensure that issues are adequately identified to the correct people and managed.
- Does the governance process include a clear approach for resolving customer disputes.
- For contracts that include cross line of business involvement, were appropriate processes established to ensure adequate representation of each line of business, and did those processes enable an effective management and oversight of the overall delivery of the solution.
- Where IBM delivery was dependent on other suppliers to the customers, were adequate governance processes implemented to enable IBM to meet its contractual requirements. This could include establishing back-to-back requirements for deliverables, provision of services and data protection requirements etc.
- Where risks and issues were recorded in contract governance meetings, were corrective actions identified and tracked, and is there evidence of ownership and effective and timely resolution.
- Where IBM delivery was impacted by other service providers to the customer, assess the adequacy of actions taken to manage those issues. Were risks identified early in the contract and processes implemented accordingly or were concerns raised once IBM delivery was impacted.
- Were the risks and issues reviewed in contract governance meetings the key risks and issues on the contract. If not, what other mechanism was used to manage key risks and issues and were they visible to senior management.
- Review the supporting management system and assess whether review meetings and supporting measurements were effective in identifying risks, documenting agreed mitigation plans, recording activities undertaken against the plan and tracked to satisfactory conclusion.
- Were issues raised to the right level of IBM management, both within the contract and to relevant geography management.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Secondary Test: Risk Management - Delivery (SE007.S)

Objective:

To provide an independent and objective assessment of the status of the project, issues, financial position, and risks. The type of reviews and the frequency of reviews can differ based upon criteria such as business line requirements, total contract value, profit levels or complexity.

Once delivery begins, risk management activities (IBM Consulting) verify that the project is meeting IBM's business objectives and that the solution will continue to satisfy the customer's requirements as defined in the contract, and risks to those objectives and customer requirements are identified and managed.

Approach:

Ongoing risk management

- Establish the mechanism to hand-over from the Solution Design process to the Solution Delivery so that risk and quality issues identified earlier on in the design stages are communicated accurately and timely to delivery stages. Look at SD005.S from Solution Design risk management.
- Confirm that a risk and issues process or other delivery assurance activities have been established and are operating as an integral part of the contract management system. Ensure that reviews include all relevant parties that impact risks on the contract (customer, IBM lead line of business, other IBM lines of business, third parties).
- Assess whether issues highlighted within the contract governance processes and management system are formally logged and tracked in the risk and issues log where appropriate. Ensure that the risk has been adequately reviewed, action plans defined including responsible owners, and tracking of the risk has been implemented including measurements?
- Review the pre-contract risk management and assurance reviews. Where risks have been identified, are these included in risk logs? Have actions been defined to mitigate those risks and measurements in place for tracking the risk or have they been allowed to materialize.
- Where risks are removed from the risk log, review the basis for closure. Are there instances where risks are re-opened, or the risk is highlighted again? Assess the adequacy in those cases of the risk containment activities established to close the risk.
- Where there have been changes in risk management or business line processes identify the process before and after, assess the impact to identify and manage risks and establish if there are any gaps where processes have been re-engineered or removed.

Periodic risk management reviews

- Review the output of risk management and assurance reviews and assess the adequacy of the reviews in identifying key risks. Assess whether key risks identified and reviewed during the audit were already identified in risk management reviews.
- Confirm that recommendations from previous reviews have been addressed, and the current status of open and ongoing issues from previous reviews.
- Validate whether key information was made available to the reviewer or management and whether significant gaps in the reviews or reviews process exist.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Secondary Test: Risk Management - Delivery (SE007.S)

(Continued)

- Review the project financial status and assess whether the root cause of issues has been identified in the review. Assess whether financial variances are the result of other aspects of the contract delivery, e.g., project management, unclear scope and whether reviews identified those risks, and put in place actions to contain and mitigate those risks.
- Is there an effective mechanism in place to identify and monitor the overall health of project. Is it a balanced score-card approach that considers several key areas of performance (e.g., customer satisfaction, financial performance, delivery against contractual agreements etc.).
- Were reviews carried out at the correct frequency and at the right level based on business line requirements.

For a sample of contracts

- Are management effectively governing risk management and quality assurance type activities.
- Consider the business line, contract value, complexity or other determining factor and establish if the mandated solutions delivery reviews related to risk and quality have occurred.
- Was IBM able to deliver the solution that was originally defined in the contract. If not, why not. Is there evidence that expectations were unclear, unrealistic, or not able to be delivered for other business or technical reasons (e.g., resource and skill availability).
- Are risks effectively identified, analyzed, and captured formally in logs or registers including customer critical success criteria.
- Are risk triggers understood and effectively identified and communicated into a risk management system.
- Do risk responses adequately mitigate risks so that the residual risk is at an acceptable level (risk appetite).
- Are owners assigned to manage risks. Is there evidence that there is clear ownership and correct ownership identification or are issues passed on.
- Are risks prioritized and managed by severity/ impact level and/or likelihood of materializing.
- Is there evidence of a universal agreement being gained to understand the root cause of risks and managing specific risks. How effective are treatments of risk, do they resolve and contain to the required level.
- Is there evidence of a risk appetite or a culture within contract teams that is not in line with the general IBM business approach.
- Is there evidence from communications throughout the contact team or business function to ensure that the right information is summarized and consolidated and reviewed by the various levels of management.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Secondary Test: Risk Management - Delivery (SE007.S)

(Continued)

- Are risk mitigations actions effectively monitored for performance and reported on to determine further decision making / actions.
- Is there evidence of missed due dates, or persistent due date slippage on addressing and closing action plans. If so, what is the cause.

Note: Consider the implications of findings from other Solution Delivery tests when assessing the effectiveness of the risk management and delivery assurance processes. Check that any errors during solution design were captured in those processes and handed over by follow-on risk management and assurance processes.

** obsolete **

For contracts using GTS Global Resource where services were transferred to a global delivery center establish that the required Cross Border Transfer (CBT) process was followed:

- a. Verify that new transfer (after implementation of the Cross Border process) with Risk Assessment - High Risk have been stopped and no customer education or access to system provided.
- b. Confirm that Risk Mitigation letter has been followed during the transition and steady state.
- c. Confirm that Risk and Issue management process have been followed for account with Risk Assessment - High Risk (workload that supported by the Delivery Locations and transfer prior the implementation of the Cross Border Process).

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Secondary Test: Systems Access and Separation of Duties (SE008.S)

Objective:

To ensure that access to IBM systems used to manage and control the CRMfS process operating on a contract and line of business is controlled in a timely manner, and to prevent unauthorized access to IBM applications and information. Appropriate management approvals should be granted and documented according to business need.

Note: controls over access to customer IT environments, applications and data would be tested under 42-00-00 Information Technology audits. Access to customer data would be tested on an exception basis on a CRMfS audit.

To ensure that separation of duties is maintained, and no single individual is performing multiple conflicting tasks to prevent unauthorized transactions. Secondary controls should be in place where conflicting tasks exist that monitor and identify exceptions.

Approach:

Obtain a list of applications used and review the process for granting approvals for access to ensure that business need is confirmed. Ensure approvals include the employee's manager, or other relevant individual, that would be aware of other access granted to that employee. Review the process for periodic validation of continued business need and ensure that access for leavers (including to other roles) is deleted in accordance with IT security timeliness requirements.

Specific items to be aware of:

- Excessive access granted to sensitive personal information and HR data, access to IBM banking and credit systems, and no secondary controls in place to mitigate. Look for individuals with multiple userids to the same system with different levels of access, or individuals with non-standard access, or where leavers/LOA/absent staff ids are still actively being used.
- Assess whether ASCA certifications are in place, and if any major application changes have occurred has subsequent ASCA reviews been made. Determine whether recent reviews (including self-assessments) identified issues and have they been addressed or are plan underway to resolve.
- Process owner, and employee's line management involvement in access authorization, revalidation, and timely removal.

Obtain a copy and review all SOD assessments including job roles assigned to employees.

Does analysis happen regularly and after significant changes.

Identify secondary controls within the SOD assessment and validate if executed as intended and that it is effective to meet the objective of the control.

Ensure that systems access for each role is an integral part of the SOD validation to identify potential conflicts.

Review whether a standard SOD analysis tool has been used.

Is SOD responsibility clearly understood. With complex matrix organizations with remote or landed staff how is an overall assessment made and are there any gaps.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Billing and Accounts Receivable (SE009.P)

Objective:

To ensure that billing is in accordance with contract terms, and that invoices and credit notes were properly justified, approved and issued on a timely basis; and that no issues remain that impact the recovery of accounts receivable.

Note: testing of billing and accounts receivable should be tested within Fulfillment - Sales Transaction Support (52-00-00) and Accounts Receivable (52-40-00) audits, and only on an exception basis tested within CRMfS. Review of billing and AR exceptions should be used to identify red flags in other CRMfS processes. Recommendations within a CRMfS audit should be to 'work with Sales Transaction Support to'.

Approach:

Billing indicators: Billing can be analyzed to detect issues in CRMfS processes. Billing adjustments or suppression activity can be an indicator that specific Solution Design and Delivery (SDD) processes are not operating as designed and can also highlight potential impact to revenue recognition.

- Confirm that the contract team has documented the expected billing profile for the contract and monitor that those terms are met. If not, was that identified and subsequently resolved. Determine the root cause.
- Ensure that the correct billing information was provided timely to those responsible for generating billing (Fulfilment, STS) and that interlocks between operations and STS operate effectively.
- Were credit notes, billing adjustments, and sales concessions correctly authorized? Assess whether supporting documentation has implications for the validity of revenue or cost recognition - in particular the timing for recognition of revenue or cost and assess whether the root cause of credit notes points to SDD issues.
- For credit notes that reverse revenue in a prior period assess whether the issue was known to the business in that prior period.
- Identify if there is any billing suppression activity and does the root cause point to issues in SDD.
- Does the billing history show any evidence of concessions or unauthorized commitments made to the customer, such as late or incomplete billing.
- Where variable consumption / resource unit billing is in place, have management or contract reviews been performed against source information to confirm accuracy?
- Are actions in place to monitor and reconcile service delivery process and adjust contract baselines for additional/reduced resource charges (ARCs, RRCs), for billing accordingly. Is the interlock between hardware and software management processes operating effectively to adjust baselines for billing. Where reliance is placed on another line of business for information impacting billing, is there an agreed process / DOUs detailing responsibilities to ensure accuracy of information? Are interlocks between service delivery and finance operating effectively to update baselines to ensure that billing, including expenses, is complete?

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Services Solution Delivery

Primary Test: Billing and Accounts Receivable (SE09.P)

(Continued)

- Is information provided to billing from a trusted source? With the potential for financial risk determine if trusted global applications that are ASCA certified were used or if local procedures and uncertified systems are operational.
- Where information supporting billing is managed by the customer, are controls in place to ensure the integrity of that information? In such instances, are application controls in place to ensure the integrity of data received from customer systems?
- For baseline billing, review the process to confirm accuracy of baselines and adjust where necessary. Have baselines been agreed with the customer. Are there any disputes relating to the validity of baselines. Has technology supporting service delivery been amended since contract signing and has the impact been included in baseline adjustments?
- Is monitoring of all contractual baselines in place, and do processes ensure that where volumes increase/decrease that ARC's and RRC's are adjusted for billing on a timely basis. Are there significant delays in making adjustments to billing and what is the root cause?

Accounts Receivable indicators: Accounts Receivable activity and analysis of universe files can also highlight potential red SDD processes and potential impact to Revenue Recognition process. Analysis of the universe (rather than transactional testing) can indicate problematic areas for further review and help with contract sample selection or specific process area risks.

- Were invoices generally paid in accordance with standard terms or do delays point to non-standard payment terms, dissatisfaction of deliverables or potentially unauthorized extended payment terms.
- For disputed invoices determine the root cause using reason codes and supporting narrative text on the dispute. Establish if it is related to SDD issues. Do solutions have performance issues, is there evidence that the solution does not match customer requirements, is there evidence that milestones are incomplete or late.
- Where there are disputes assigned to marketing teams were they working actively to resolve the issues and how. Do issues appear resolved with concessions or future unauthorized commitments.
- Is there evidence of aged debt in conjunctions with / or suppressed late payment fees. If so, is there evidence that this could be withheld payments due to SDD issues or unauthorized payment terms (e.g., payment period being extended).
- Were there any write-offs or concession adjustments (look for the write-off reason code or other general ledger adjustment) made in the system that indicate issue associated to SDD. Identify situations where marketing teams have directed write-offs or concessions rather than AR decision.
- Are customer entities, accounts, or invoices in pre-legal or legal status with issues relating to SDD. Review the background, including communications between IBM and the customer, to determine the cause.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Data Security and Privacy

Objective:

Evaluate whether data, including personal information (PI) and sensitive personal information (SPI) is appropriately secured and protected.

Risk/Exposure:

Failure to protect PI and SPI could enable fraudulent activity, which could result in dissatisfied employees, damage to employees and to IBM's image, and expose IBM to risk of litigation.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Data Security and Privacy

Primary Test: Data Protection and Privacy (DS001.P)

Objective:

Evaluate whether data, including personal information (PI) and sensitive personal information (SPI) is appropriately secured and protected.

Approach:

Understand if the contract(s) with customer requires any control on themes related to data privacy and data security/protection.

Evaluate if PI and SPI are protected so that only the minimum amount of information necessary to execute the process is shared with those who perform the process.

Data Privacy

1. Review the customer's definition of data privacy and documentation requirements to protect customer and end user information. Were all data privacy requirements agreed in the contract fulfilled? (i.e., criminal background verification, etc.) Were there proper controls in place to ensure that customer's information was secured?
2. Review contracts with vendors or contractor employees. How is data privacy assured? How does IBM monitor vendor protection of data?
3. If applicable, review if General Data Protection Regulation (GDPR) requirements were deployed, controls were in place, monitored and reported.

Useful links (refer to reference material)

General Data Protection Regulation (GDPR)

GDPR Readiness Community

Compliance, Audit, Risk Mitigation and Management

Test: Privacy – Note the privacy tests are updated by CAAS every 6 months and the most current audit programs for IBM as a Controller and IBM as a Processor should be referenced during the audit.

Objective:

Privacy laws impose requirements on how IBM and its Clients process Personal Information (PI). The Global Privacy Control Assessment Questions are composed with two key areas of the business to help address compliance with these various high-level laws.

- **IBM as a Controller** – The Enterprise Privacy Baseline – Controller establishes the high-level requirements that IBM must meet when acting as a Controller of Personal Information (PI). That is, PI that IBM collects and processes for its own purposes, such as Business Contact Information that IBM holds on its existing or potential customers or suppliers.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Data Security and Privacy

Primary Test: Data Protection and Privacy (DS001.P)

(Continued)

- **IBM as a Processor** – The Enterprise Privacy Baseline – Processor establishes the principles and requirements that IBM must meet when acting as a Processor of Client Personal Information (PI) and applies to all products, offerings, and services (including their components) that IBM provides to Clients, regardless of whether IBM develops or re-brands them, or provides them for a fee or on a beta, trial, or no-charge basis.

Testing note:

Processing client-controlled PI is primarily covered by the Application Management – External audit program, which should be considered when evaluating the extent to audit processing client-controlled PI as part of a CRM for Services audit. For example, testing compliance to client contract T&Cs on PI is one element that would be in scope for a CRM for Services audit. However, testing execution of the service delivery processes, TOMs, etc, would be more applicable to an AMe audit.

Approach:

Review the Data Privacy responses provided from the Initial Data Request. Determine if IBM is Controller or Processor of PI/SPI data in the process.

Testing:

IBM as Controller

If IBM is Controller, request a risk-based sample of Global Privacy Assessments for the processes and/or assets. Complete the testing from the following link: <https://ibm.ent.box.com/file/1198251968388>

IBM as Processor

If IBM is Processor, select a sample of products/offerings/services provided for the in-scope activities. Complete the testing from the following link: <https://ibm.ent.box.com/file/1198263127550>

IBM as both Controller and Processor

Test either IBM as Controller or Processor, do not test both. Consider the volume and sensitivity of the controller and processor data and select the higher risk.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Financial Management

Objective:

Are revenue and cost recognized accurately? Are billings and invoice adjustments done timely and accurately? Are claims and concessions following the applicable process?

Risk/Exposure:

- Untimely recognition of revenue and cost
- Delayed cash flow
- Financial misstatements
- Unapproved loss of profitability
- Customer dissatisfaction

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Financial Management

Primary Test: Financial Management (FM001.P)

Objective:

Was contract performance properly and accurately monitored and were results reported? Were there management controls to ensure that contract financials, including budgets/forecasts, are accurate, properly maintained and supported, and consistent with pricing case? Were there significant backlog adjustments that indicate other issues in financial management [Note: Signings and backlog is a separate audit program, and the overall process tested within a specific S&B audit, not CRM for Services audit program.]

Approach:

Forecasting and reporting

- Access whether information for financial management and oversight is complete, accurate and timely (the right information and the right time is provided for sound decision making). Establish that financial data is from a trusted source.
- Review the process for accurate and timely forecasting.
- Review the process for tracking from price case actuals. Are adjustments to the base pricing case made on a timely basis to enable a like for like comparison between pricing case and actual to be maintained. (e.g., for additional resource charges (ARCs), reduced resource charges (RRCs), requests for service (RFSs) and contract changes.
- Where actuals deviate from the planned price case, has that been identified and tracked.
- Is effective gap / variance analysis performed to identify the root cause of financial deviations and are there appropriate action plans with an achievable roadmap in place to bring back on track.
- Confirm that the financial impacts of risks and issues in the contract are accurately reflected in the forecast. Is delivery deviating from the planned price case and is that being tracked. For example, are delays in transition and transformation (T&T) leading to future cost and gross profit (GP) impacts or are there missed service delivery requiring additional costs through to the end of the contract.
- Ensure that effective interlocks between the various key drivers of cost and service levels on the contract, and the financial management processes are in place. Assess whether information is passed regularly and timely to the finance and accounting teams. Is information sufficient and accurate to enable changes in the forecast to be quantified and revenue considerations to be evaluated accurately.
- Where IBM has an arrangement for open book reporting / accounting, review the contract requirements to share information with the customer. Assess the controls in place to validate and review information shared with the customer.

Project control / portfolio analysis

- Is there a control process which highlights projects which do not report results within expected financial returns (pricing case or revised forecast)? Are all contracts included in such an analysis at individual contract level?
- Consider over- and underperformance of gross profit when compared with typical gross margins, or the pricing case, and whether root cause of issues has been identified and actions / route map put in place to manage risks.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Financial Management

Primary Test: Financial Management (FM001.P)

(Continued)

- Has a periodic review of under-performing / troubled projects been implemented? Review the process for this. Are recovery dates for contracts defined, and met? Have recovery plans including deliverables, milestones and measurements been developed and met?
- Is spending and other costs monitored to the price case.
- Are cost over-runs promptly forecast, identified, and reported to management and accounting team.
- Financial tracking was performed with sufficient granularity to understand variances and implement and monitor effective action plans.
- If there were any adjustments to the pricing case for contract changes and request for services, they were made on a timely basis and other business functions (such as accounting) were informed to ensure compliance to financial reporting requirements.
- Establish if there were there any contractual requirements relating to service optimization and improvements that would results in payouts. Did they occur as required, were they calculated correctly and based of valid supported evidence and have they been reported.
- Have the required book to bill (B2B) reconciliations been completely regularly and if there were any variances, were they identified, analyzed, and resolved.
- If the contract is closed, establish if the closure process and associated reconciliations were performed to ensure that all revenue and costs have been correctly identified and recorded, and that all invoices or credits have been issued to the customer.

Other income and/or other charges

- Query the general ledger to verify whether other income or other charges were recorded and properly accounted for.
- Determine how the organization records other income and charges in the general ledger. For example, some organizations may record other income and/or other charges by department number, others by contract number.
- Assess the impact on GP reporting of correctly reporting charges above / below gross profit.
- Was other income and/or other charges recorded in the correct time period?
- Are there any instances where contract expenses are included in other charges leading to GP being overstated?
- All costs were allocated accurately to the contract. This includes labor costs incurred by local and international resources.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Financial Management

Primary Test: Financial Management (FM001.P)

(Continued)

Transition and Transformation

Where contracts contain T&T establish if:

- The appropriate T&T financial control system was in place to manage deferral amounts within approved amounts.
- Cost case assumptions on T&T were based on established guidelines and accounting methodology and practices.
- Where T&T overruns occur establish the root cause and also determine the financial impact. Identify if changes in performance and delivery were reported to finance teams.
- Analyze project reviews to establish if and when potential risks and issues in T&T are being reported. Establish if that information is relayed timely to finance and accounting teams.

Capital allocations

- Ensure deliverables included in the contract which require capital expenditure, have been planned and requested in order to meet the contractual requirement.

Claims / claims avoidance

- Were any services / costs not charged to the customer in accordance with the contract? Were appropriate approvals in place for concessions?
- Were concessions made as part of a contract negotiation which should have been treated as a claim?
- Were commitments to provide a concession made to the customer prior to IBM approvals?
- Was root cause analysis of the issue conducted to minimize the risk of further lost revenue?
- Is there evidence of management oversight and review in case of negative trends?

(FOR THE US ONLY) Extended customer assignments within in the employee's home country:

- Review the process for tracking and monitoring the duration of employee assignments on the project.
- Assess whether the pricing of the contract took into consideration long-term assignments and the potential for additional costs associated with tax assistance.
- Ensure employee expenses were taxed from the point in time at which it was known their remote assignment would exceed the permitted duration.
- Validate that breaks in continuous service were managed in accordance with the guidelines and did not trigger additional income tax liability. Ensure that breaks in service were not being initiated and recorded with the explicit intent to re-set the allowed duration (e.g., US one-year rule).

(FOR THE US ONLY) Primary tax homes:

- Review the process for managing apartment approvals and long-term hotel stays. Validate that primary tax home guidance was provided to employees approved for such expenses.
- Ensure controls were in place to validate that employees who had apartment and long-term hotel expenses were in compliance with primary tax home guidelines.
- Perform appropriate testing to validate employee compliance with primary tax home guidelines.
- For instances in which employees were not compliant with primary tax home guidelines, had the reimbursement of remote living expenses been discontinued, and financial reporting corrected?
- Ensure inappropriate remote living expense reimbursements have not been billed to customers, resulting in a financial statement impact.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Financial Management

Primary Test: Revenue Recognition (FM002.P)

Objective:

To ensure that revenue recognition was in accordance with generally accepted accounting principles and IBM practices.

Note: The auditor should review in detail the terms and conditions of contracts selected in sampling to ensure that the revenue is recognized consistent with contractual commitments.

Approach:

The Initial Data Request (IDR) requires a report of the revenue recognized by the business unit or project over the audit period. Reconcile the data provided with the reported results for the unit / project. When assessing the risks relating to accounting entries, analysis of automated entries versus those that are subject to change and/or have potential for manipulation should be assessed (e.g., certain software revenue recognition may be fixed on a transaction regardless of contract terms, geography), to ensure focus is placed on highest risk revenue in testing. Consideration in sample selection should be focused on fixed price rather than time and materials contract elements, on revenue subject to changes in volumes delivered rather than fixed charges.

When selecting transactions for review focus should be on the following:

- Entries processed at quarter ends and year-end.
- High value entries (especially manual entries).
- Revenue recognized based on triggers or input from groups outside the accounting function. For example, project office and project management teams in cases of percentage of completion (POC).
- Revenue adjustments processed at the Headquarter level.
- Material balance sheet adjustments including cost deferrals, high value accruals - including transition and transformation deferrals.
- Significant claims and timing of provisions.

Ensure compliance with the relevant accounting instructions and guidelines. For the selected sample, verify that services revenue has been recorded when earned. In general, revenue should be recognized and recorded when the following has taken place:

- A firm customer order exists with scope of work defined and a signed contract is in place.
- Evidence that services were rendered.
- Collection is reasonably assured.
- The fees for the services are fixed and determinable.

Key attributes to consider in the review of revenue recognition should include:

- Revenue is recognized in the correct accounting period, and that revenue and cost matching principles were adhered to.
- Prior quarter reversals are supported and justified. Also establish that the basis for the original revenue recognition was valid at the time of the original booking.
- Cost accruals were booked in the correct accounting period.
- Management review / clip levels were adhered to, and adequate separation of duties exists for the processing of manual accounting entries transactions.
- Have pass-through / cost recoveries been assessed for revenue recognition implications.
- Consideration should be given to information that indicate issues with revenue recognition (e.g., downgrading of credit rating indicating collection issues, claims, disputes, or legal referrals).

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Financial Management

Primary Test: Revenue Recognition (FM002.P)

(Continued)

- Manual entries: assess whether manual entries were required to correct other issues and validate which aspect of a business-as-usual process did not ensure correct revenue recognition. Where manual entries are the basis of transaction, review the adequacy of controls to ensure that they are correctly calculated, including a review of supporting documentation. Assess whether those transactions are processed on a timely basis with supporting backup, and where estimates are used, assess the validity of those estimates.
- Reversal of revenue: assess whether any conditions exist that require the reversal of revenue previously booked. Are controls in place to monitor potential reversals, e.g., monitoring of billing on services contracts to reverse revenue for unbilled in accordance with Accounting policy.
- For non-standard transactions, determine what accounting reviews would be necessary and establish if they occurred properly by the appropriate subject matter experts and were performed at the correct time (e.g., global accounting checklist).
- Establish if revenue or cost transfers from other lines of business were recorded timely.
- Determine if costs were allocated accurately to the contract such as labor costs incurred by local and international resources.
- Expenses and costs were reported accurately as expense or cost category.
- Reconciliations such as book to bill were completed accurately and timely and discrepancies were investigated and resolved.
- Revenue and costs booked associated to percentage of completion (POC) contracts was done so in accordance with accounting guidance.
- Where gross profits have fluctuated with POC contracts, establish the cause, and determine if continued POC methodology is appropriate.
- Establish if provisions have been made when required for contractual non-performance, missed SLAs, missed milestones, penalties, termination charges or non-standard contractual terms.

For contracts that include transition and transformation (T&T) assess whether:

- Deliverables constitute assets equal to the value of deferrals to the balance sheet. In cases where significant cost overruns have occurred, disputes are in place over the quality of deliverables, or the scope of T&T has significantly changed since approvals for T&T deferrals assess whether Accounting review has been performed and information made available to Accounting to enable an accurate assessment of the value of the asset to be performed.
- T&T costs are separable from other charges to the customer, and management has reviewed that only work performed on T&T is logged as such.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Financial Management

Primary Test: Revenue Recognition (FM002.P)

(Continued)

- Review the release of deferred T&T revenue and costs to ensure that write backs are consistent with the period when the benefit of T&T is received.
- Assess the adequacy of controls to validate the accuracy, and timing of Additional Resource Charge (ARCs) and Reduced Resource Credit (RRCs) and their impact on revenue recognition.
- Where revenue is based on labor recording, assess whether subsequent reversals to records were made, and the potential for individuals other than those performing the work to manipulate labor recording.
- Where work at risk / pre-contract work is an integral part of contract management, review controls in place to ensure revenue is only recognized on signing of a contract.
- For contracts with significant OEM content, assess whether revenue has been booked as gross or net. For revenue recorded as gross, assess whether the criteria for revenue recognition have been met, split across primary and secondary indicators of where IBM is the primary obligor. Assess whether such transactions have been assessed in accordance with clip levels by the contract team / F&P / Accounting as required. This aspect of testing is often judgmental by nature, and should include validation of indicators that include: whether IBM was acting as an agent rather than the primary obligor; balance of core value-add services versus non-core value-add services; which party bears the risk of inventory; value add provided by third parties; contact between IBM's end customer and IBM's suppliers and the end customer perception of which party will address issues with IBM's service.
- For hardware shipments, review for reasonableness of dates (shipment dates, billing date, revenue recognition month, accounts receivable collection date, supplier selection dates, and payment dates) as indicators for abnormal transactions which require further analysis. For example, review: supplier selection versus vendor shipment date and shipment versus billing versus revenue recognition versus accounts receivable collection date.
- For Strategic Partner funding was revenue recognized or cost offset performed per accounting review and advice.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Labor Claiming

Objective:

To ensure that labor claiming is recorded in a timely and accurate manner. To ensure that labor charges are assigned to the correct work items, rates and bands agree to employee records, and hours are validated against applicable time recording documents and internal interlock agreements.

Risk/Exposure:

Inaccurate clocking of labor hours could result in significant over or under recovery of cost. In addition, incorrect assignment of labor charges to work items and inaccurate rates and band levels compared to employee records could result in financial misstatement.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROLASSESSMENT QUESTIONS

Key Control Question: 5. Labor Claiming

Primary Test: Labor Claiming and Work at Risk (LC001.P)

Objective:

To ensure that labor claiming is accurate and in accordance with IBM and line of business guidelines for full labor recording, and controls are in place to ensure the integrity of billing and measurements.

To ensure that work is not commenced prior to customer commitment or without following the required IBM work at risk / pre-contract work processes.

It should be noted that some business units employ pre-contract work as a sales enablement tool demonstrating IBM's capabilities. In such cases, and in all other cases of work undertaken outside of a signed contract, an assessment should be made of how the business monitors to identify for unapproved work at risk / pre-contract work, how liabilities are limited or contained with customers to protect IBM, and how violations are dealt with. Business Conduct Guidelines (BCG) specifically requires employees to record hours worked accurately, any instructions or decisions to do otherwise violates BCG and should be reported for further investigation.

Approach:

As part of the process review request details on the management system governing time recording and labor claiming (including detail of specific codes usage and how this relates to work). In addition, ensure local labor laws and band level / employee type limitations regarding overtime re addressed.

Labor claiming and work at risk / pre-contract work are unique area's that pose different risks. For example, inaccurate labor claiming may result in missed or inaccurate billing and inaccurate utilization recording. Work at risk or pre-contract work may result in legal dispute including unlimited financial penalties if not properly controlled. There are also links between these areas. Inaccurate labor records maybe deliberately hiding and circumventing work at risk or pre-contract work processes. Unauthorized work at risk and pre-contract work may be hiding other issues, such as unapproved concessions or solution design and delivery issues. When reviewing these areas consider how and if erroneous activities can be effectively identified and monitored by management and also the root cause, why were hours recorded incorrectly, why did we provide 'free' services.

Labor claiming testing is looking at the accuracy and timeliness of recording work effort. Is the right type of work being recorded to the right project by the right person at the right time to allow correct billing and utilization of staff?

Work at risk and pre-contract work needs to be properly identified and managed in advance of being performed to ensure IBMs liability can be limited. Unapproved work on customers live systems could have significant financial, legal, and reputational damage in case of fault for example. Such work effort can also result in lost revenue or unrecoverable costs if not properly controlled. Test to see that work is controlled with established and adhered to processes that ensure work is properly authorized and agreed prior to commencement, that the scope and the duration or work does not exceed those agreements, and that IBM is able to recoup effort associated with effective and met conversion rate target.

Review labor claiming records and identify exception situations including:

- Significant transfers between projects and contracts, indicating potential work at risk.
- Net negative hours for employees on specific projects.
- Excessive labor claiming (review average claiming by banding).

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Labor Claiming

Primary Test: Labor Claiming and Work at Risk (LC001.P)

(Continued)

- Hours transferred between billable and non-billable activity codes including the transfer from or to vacation and education time to billable.
- Significant gaps between when labor was claimed, and the date claimed for.
- Codes where expenses have been incurred but no labor claimed? Are there any individuals with expenses and no hours claimed?
- Claiming by individuals on behalf of others, it may be possible for a project manager / administrator to claim on behalf of someone else, however excessive time claimed could indicate inappropriate claiming.
- Time recorded against contracts which are re-opened during the audit period so that additional hours can be recorded. This could identify that revenue had been prematurely recognized against the original contract.
- Manipulation of under-performing contracts i.e., when a contract under-performs, are spare resource booked against it to make other contracts and utilization look better.

Review reports detailing with project/RFS revenue and cost for situations that indicate issues in labor claiming. Identify projects with no costs, no revenue or excessive costs that may indicate:

- Hours recorded against expired or terminated contracts.
- Billable hours not claimed against a billable claim code, with potential missed billing.
- Hours worked against a project but not claimed or hours claimed but not worked.

Review a sample of contract changes and requests for service (as per test SE005.P) to identify work at risk/pre-contract work:

- Review the management system in place to identify work started before contract. Assess the adequacy of that process where instances of undetected work at risk or pre-contract work existed.
- Review all correspondence for the RFS/contract change to determine if there is any indication of work being carried out prior to contract signature.
- For approved work at risk and authorized pre-contract work, check for required approvals prior to commencement of work. This should include communications and agreements with the customer to limit IBMs liability, internal business lines approvals with a clear understanding of scope (both nature of work and period of work expected).
- Review whether work continued after approvals expired or cost limits had been exceeded. Was that identified and mitigated, if not why.
- Determine if there was a mechanism to ensure IBM is able to eventually identify the work performed and bill.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Labor Claiming

Primary Test: Labor Claiming and Work at Risk (LC001.P)

(Continued)

- Is there evidence that pre-contract work or work at risk is abused with a lack of conversion to agreed contract, thus providing an unapproved concession for the customer.
- Assess whether controls were in place to monitor for the expiry of work at risk approvals or pre-contract work, including approaching limits.
- Assess the impact of work at risk and pre-contract work on revenue recognition without a contract.
- Review where subcontractors have started work without a signed contract in place, or prior to IBM's contract with the end customer (note, subcontractors can work at their own risk, establish if IBM requested them to start work, and potential for procurement bypass).
- Determine if the amount of work at risk or pre-contract work is reported accurately and is in line with any set business targets. A permitted amount maybe provided by the business as part of negotiation activities, however, how are conversion rates from pre-contract work to signed contracts monitored to ensure the expected percentage is subsequently billed on to customers.
- Establish if the risk of IBMs liability and potential penalties are limited prior to work performed before contract by method of a customer commitment letter, letter of intent, or other legally reviewed and approved document that limits liability. Assess the type of work performed prior to contract agreement to establish risk.
- Does the scope of work performed match the basis for business approvals and agreements that limited IBMs ability. There is potentially more risk where engineers access customer live equipment or deploy software code on customer systems versus providing education or consulting services. Performing services work that goes beyond approved and agreed activities could also render agreements to limit liability as invalid.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Resource Management

Objective:

To validate that there adequate on-boarding and off-boarding process in place to ensure all customer requirements are being met, that staff understands how to protect customer data, that staff members have been properly vetted and trained, and that team members remove all customer data from their possession when leaving an account.

To understand if there any IBM or local enterprise risks identified related to the IBM services provided within the contracts and how those risks were tracked and managed.

Risk/Exposure:

Non-adherence to customer requirements and internal procedures for staffing engagements could lead to legal exposures, damage to IBM's image, misuse of customer data, disrupt customer business and lead to customer dissatisfaction.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES

AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Resource Management

Primary Test: Resource Management (RM001.P)

Objective:

Assess the alignment between resource planning, price commitments, and the amount of resource / cost used to deliver the work. Validate that investments (i.e., hiring, sub-k) are supported by capacity analysis and/or unique skill requirements and approved by authorized individuals. Assess management's ability to identify and address differences between resource plans and delivered results. Ensure risks were assessed and proper agreements were in place when the use of subcontractors or vendors was needed. This scope is part of the Labor Claiming test scope.

Approach:

Potential indicators prompting higher focus of Resource Efficiency in the audit scope:

- By-passes of WFM processes – hiring managers onboard new hires or sub-k (new and renewal) outside of approval processes. Reconcile hires to approvals in tool used to request new hires and identify transactions which were not approved.
- Demand requests are not in line with pricing case – review band profile, sourcing mix (global delivery vs. domestic) in the approved price case. For accounts which had signings in the last year, select those which are not delivering to their price case (without contingency).
- Bid metrics exception. (Note approval is no longer required for bid metrics exceptions.) For accounts which had signings in the last year, select those opportunities which exceeded the bid metrics targets (avg band > bid metrics, GIC % < bid metrics).
- Experienced Professional Hire (excluding Grad hires / Associates / Students / Interns) request on-board date is more than 30 days before start date of project (scope includes Fixed Term Hires, LTS, Regs).
- Root cause of new joiners without billable utilization (Grad hires and Go-to-Market out of scope, focus on delivery).
- Project margin in approval tool is close to delivered margin (sub-ks and hires).
- Exceeding demand interlock authorized net headcount – was there a management system to monitor adds and exits (hiring, attrition, etc.). If they exceed authorized net headcount, was it approved according to delegation.
- Ineffective bench management – no visible demonstration of management actions to address practitioners on the bench.
- Promoted individuals are not redeployed to opportunities at corresponding higher bill rate.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Resource Management

Primary Test: Resource Management (RM001.P)

(Continued)

Services Solution Design - Resource Management

1. Was a staffing risk assessment completed, if required?
2. Was a risk management review conducted and artifacts loaded to the Deal Risk Assessment (DRA)/Deal Compliance Approval (DCA) or current repository?
3. Were recommendations specific to staffing identified and addressed prior to contract signature? Was a pricing release review performed by the Account Team with the appropriate Geo and Market approvals, consistent with pricing delegation, including any required Solution Release Review (SRR) and Pricing Release Review (PRR)?
4. Was pricing and staff ability approved in DCA or current tool?
5. Was the project forecast submitted into the forecasting tool (IPPF) and approved? Was the deviation to price case identified and was the use of contingency approved?
6. Were required approvals for hires / sub-ks obtained in approved local tool?

Services Solution Delivery - Resource Management

1. Compare staffing plan to resource assumptions in the price case to resources levels used to deliver the work – are all three in alignment? If not in alignment, is this visible in the project forecasting tool (IPPF) and is management pursuing action plans to recover significant differences.
2. Was the staffing plan submitted to Workforce Management in alignment with pricing assumptions, including contingency?
3. Post contract signature, examine changes to staffing plans and/or open seats and identify whether a significant deviation from price case potentially required a new solution and/or PCR and was it proposed.
4. If use of subcontractors or vendors was required to deliver the service, were customer requirements followed? Did employees appropriately claim their labor, in order to correctly charge the customer?
5. If subcontractors were used, confirm that a seat was opened to allow for a match of available practitioners with the required JRS. If there were practitioners on the bench with similar skills noted in the approval tool request, was a note included on why they were not used – for example, niche skill requirements, security requirements, language requirements, location requirements, non-strategic skill.
6. Review process for claiming hours to the contract. Were hours claimed correctly and in a timely manner? Were hours claimed consistent with the commitments identified in the contract for the services to be performed?
7. Review labor claim policy and analyze claim categories to ensure consistency.
8. Review whether contractor and/or overtime hours were aligned with labor laws.

Education and Recruitment

1. Review training and education processes for new and existing employees, subcontractors, and vendors. Were there sufficient resources to execute contractual obligations? Were the skills of the workforce adequate for all job tasks defined?
2. Review contract to identify requirements for Global Resourcing. If a requirement exists, were reviews, controls, and oversight in place to ensure IBM is able to meet contract commitments?
3. Did management establish measurements to determine the effectiveness of recruitment activities? Ensure that all supporting documentation is held according to legal and company requirements.

CUSTOMER RELATIONSHIP MANAGEMENT FOR SERVICES AUDIT PROGRAM #60-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Resource Management

Primary Test: Resource Management (RM001.P)

(Continued)

Vendors

1. Was there evidence of procurement involvement for subcontractor vendor quotes, if use of subcontractor was identified during solutioning / pricing? Were there agreements in place for those subcontractors or vendors which support IBM in delivering business process services? Provide documentation that subcontractor approval went through CSA/SP@I. If not processed through CSA or SP&I provide documentation from Procurement that they were engaged.
2. Did procurement engage with Supplier to support PO issuance? Was there an invoice without a PO (procurement by-pass).
3. Was there a periodic control of subcontractor or vendor labor hours completed executed? Did this control assure that subcontractor or vendor services started after purchase order approval?
4. Where customer required non-disclosure or confidential agreements, ensure that vendors were included under such, and had signed similar clauses with IBM Procurement.